

Integrated Municipal Information System (IMIS)

Web Application

User Manual



Supported by



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1 Introduction

1.1 Purpose

The purpose of the document is to instruct users of IMIS on the functionalities of web interface of IMIS. It details step by step procedure of all implemented functionalities within IMIS.

2 IMIS Login

On accessing IMIS web application on a browser, the user is asked for their login credentials. IMIS web application can be accessed with any browser however the developers are recommended to use Chrome browser.

- Enter Username and Password in textbox.
- Click on Login button
- Use **Remember Me** checkbox to store the login credentials

The screenshot shows the login interface for the Jhenaidah Municipality Integrated Municipal Information System (IMIS). The header features the municipality's logo and name in Nepali and English. The login form includes fields for Username and Password, a 'Remember Me' checkbox, a 'LOGIN' button, and a 'Forgot Your Password?' link. A red circle highlights the text 'Developed for Chrome Browser' at the bottom of the page. Two orange arrows point to the Username and Password fields with labels 'Enter username' and 'Enter password' respectively.

Fig. 1: Login Screen

3 Sidebar Menu

After logged in the page, a dashboard page is displayed (the first item in the side bar menu). The side bar can be expanded and collapsed. There are different eight modules on the side bar.

1. Dashboard
2. Building Info Management
3. Utility Info Management
4. Data Export
5. Data Import
6. Settings
7. View Maps

Some of the modules can be further expanded to reveal a child menu. Depending on user credentials, only specific modules on the side menu will be available to the specific users.

4 Dashboard

A dashboard is an information management tool that visually tracks and displays many key performance indicators and key information about the IMIS. With the dashboard the user can be get a graphical synopsis about IMIS and get many key information about IMIS at a quick glance. There are five key indicators and 11 different types of graphs in the dashboard. The dashboard is linked with the IMIS database such that all key indicators and graphs are constantly updated. For example, if a newly created building is added in the IMIS, the key indicator Structure Building will be automatically updated.

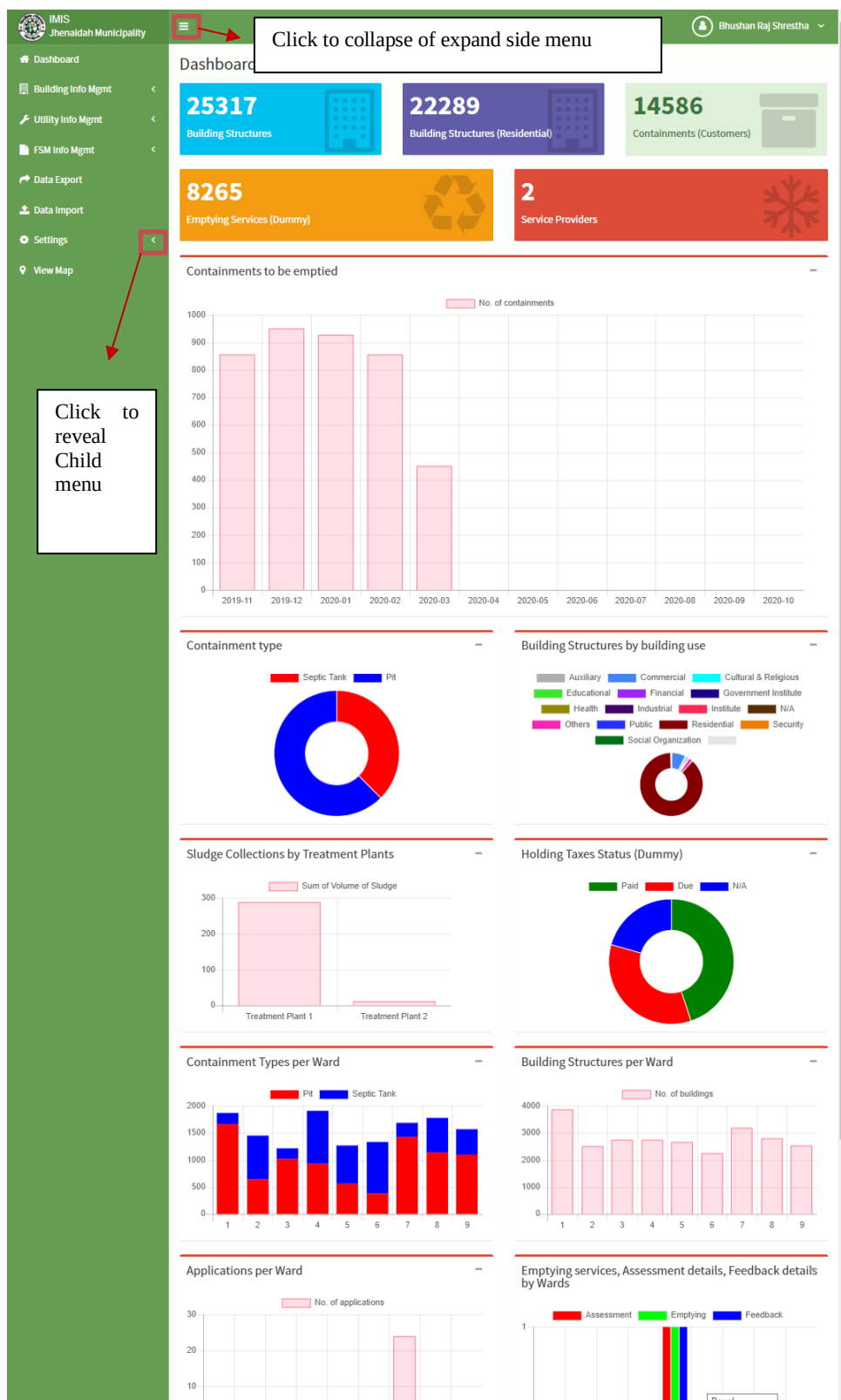


Fig. 2: Dashboard

5 Building Information Management

This is a module dedicated to support mapping and maintaining building database.

5.1 Building Structures

- Navigate to Building Info Management > Building Structures

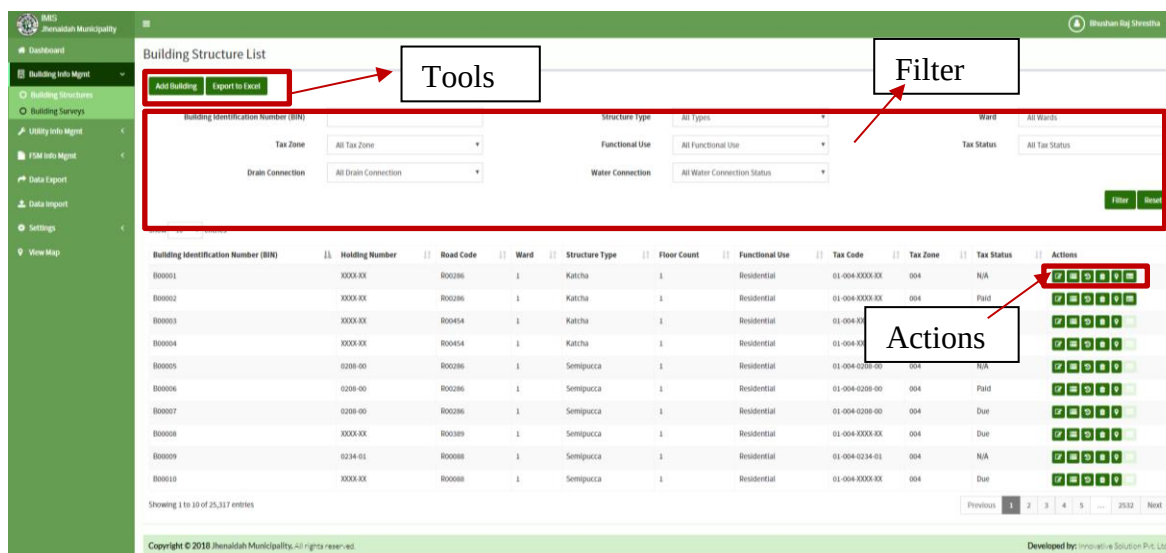


Fig. 3: Building Structure List

Only some of the important attributes are shown in the list. Attributes shown can be sorted in ascending/descending order. The user can search for a particular building according to its BIN number. There are different filters available that enables user to filter the building list according to their need. E.g. if only building within a specific ward is needed, ward filter can be used. There are different Actions buttons available for each record. The user has a flexibility to choose how many records (10, 25, 50 & 100) are displayed on a page.

5.1.1 Filter

- Enter a specific Building Identification Number/Holding Number/Tax Code or Road Code and click on **Filter** button.
- Choose an appropriate Filter as desired, select its appropriate value and click on **Filter** button. E.g. Structure Type > Pucca Building and Ward > Ward 1. To remove filter, click **Reset**.

Building Structure List

[Add Building](#)
[Export to Excel](#)

Building Identification Number (BIN)
 Tax Zone
 Holding Number
 Drain Connection

Structure Type Ward
 Functional Use Tax Status
 Tax Code Road Code
 Water Connection

[Filter](#)

Show entries

Building Identification Number (BIN)	Holding Number	Road Code	Ward	Structure Type	Floor Count	Functional Use	Tax Code	Tax Zone	Tax Status	Actions
B00029	0168-01	R00377	1	Pucca	1	Residential	01-004-0168-01	004	Due	     
B00045	0175-00	R00346	1	Pucca	1	Residential	01-004-0175-00	004	Paid	     
B00086	0186-01	R00376	1	Pucca	1	Residential	01-004-0186-01	004	Paid	     
B00087	0168-00	R00377	1	Pucca	1	Residential	01-004-0168-00	004	N/A	     
B00088	0177-00	R00352	1	Pucca	1	Residential	01-004-0177-00	004	N/A	     
B00114	0203-00	R01889	1	Pucca	1	Residential	01-004-0203-00	004	N/A	     
B00115	0143-01	R00266	1	Pucca	1	Residential	01-004-0143-01	004	Paid	     

Fig. 4: Using Filter to view specific buildings

5.1.2 Actions

There are six different Actions available for building. Use **Details** Action when the user wants to just view attributes of the building. Use **Edit** Action when there a need of modifying any particular attribute the building. **Water Payment** Action is only enabled to buildings that are connected to water network.

Table 1: Actions for building data

ICON	Name	Task that will be performed
	Edit	Opens form to edit the building details
	Detail	Shows detail information about the building
	History	Shows information about log history of edits
	Delete	Delete building data
	Map	Shows spatial location of the building on a map
	Water Supply Payment	Shows water payment status for buildings that are connected to water network

Action: Edit

- Click on the **Edit** button in the Action column of the Building Structure List
- In Edit Building Structure page, enter empty fields or change/modify already added fields as necessary
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Building Structure List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Building Structure List page

Edit Building Structure

Building Identification Number (BIN): B00001

Holding Number	XXXX-XX
Tax Code	01-004-XXXX-XX
Road Code	R00286
Ward	1
Structure Type	Katcha
Floor Count	1
Estimated Area of the Building (sqm)	30.1617
Functional Use of Building	Residential
Use Categories of Building	Use Categories of Building
Office or Business Name in Building	Office or Business Name In Building
BIN of Main Building	BIN of Main Building
Surveyed Date	Surveyed Date
Draincode	D00020
Drain Connected Date	2019-08-01 00:00:00
Drain Connection	Drain Connection
Water Connection	Yes
Water Supply Customer ID	TEST CUSTOMER ID
Water Connection Date	2019-07-09
Customer	Customer Name
Contact	Customer Contact

Back to List
Save

Fig. 5: Building Action- Edit

Action: Detail

- Click on the **Detail** button in the Action column of the Building list
- Explore attributes of buildings as desired
- Click on **Back to List** to go to Building Structure List page

Building Structure Details

[Back to List](#)

Building Identification Number (BIN)	B00001
Holding Number	XXXX-XX
Containments	C03666
Tax Code	01-004-XXXX-XX
Road Code	R00286
Ward	1
Structure Type	Katcha
Floor Count	1
Estimated Area of the Building (sqm)	30.1617
Functional Use of Building	Residential
Use Categories of Building	
Office or Business Name in Building	
BIN of Main Building	
Surveyed Date	
Draincode	D00020
Drain Connected Dateable	2019-08-01 00:00:00
Drain Connection	
Water Connection	Yes
Water Supply Customer ID	TEST CUSTOMER ID
Water Connection Date	2019-07-09
Customer	
Contact	

Fig. 6: Building Action- Detail

Action: History

- Click on the **History** button in the Action column of the Building List
- View the changes made to attributes of the building
- Click on **Back to List** to go to Building Structure List page

Building Structure History

[Back to List](#)

- Bhushan Raj Shrestha changed water_connection_date from 2019-07-09 00:00:00 to 2019-07-09 on 2019-10-01 08:35:05
- Karan Bajracharya changed water_connection_date from to 07-09-2019 on 2019-07-05 12:36:49
- Karan Bajracharya changed water_supply_customer_ID from to TEST CUSTOMER ID on 2019-07-05 12:36:49
- Karan Bajracharya changed water_connection from to Yes on 2019-07-05 12:36:49
- Karan Bajracharya changed drain_connected_date from to 08-01-2019 on 2019-07-05 12:36:49
- Karan Bajracharya changed draincode from to D00020 on 2019-07-05 12:36:49

Fig. 7: Building Action: History

Action: Delete

- Click on the **Delete** button in the Action column of the Building list.
- It will display confirmation dialog
- Click **OK** button in the confirmation dialog to delete the building

IMIS Jhenaidah Municipality

192.168.124.40 says
Are you sure?

Building Structure List

[Add Building](#) [Export to Excel](#)

Building Identification Number (BIN) Structure Type Ward [Filter](#)

Show entries

Building Identification Number (BIN)	Holding Number	Road Code	Ward	Structure Type	Floor Count	Functional Use	Actions
AK20002						N/A	Edit Delete Map
AK20003							Edit Delete Map
AK20005	11	33	44	Under Construction	55	77	Edit Delete Map
AK20055							Edit Delete Map
B00001	XXXX-XX	R00286	1	Katcha	1	Residential	Edit Delete Map
B00002	XXXX-XX	R00286	1	Katcha	1	Residential	Edit Delete Map

Fig. 8: Building Action: Delete

Action: Map

- Click on **Map** button in the Action column of the Building list. A map will open, showing a spatial location of the building with a marker

- To go back to Building Structure List page, either click Back button on the browser or navigate to Building Info Management > Building Structures

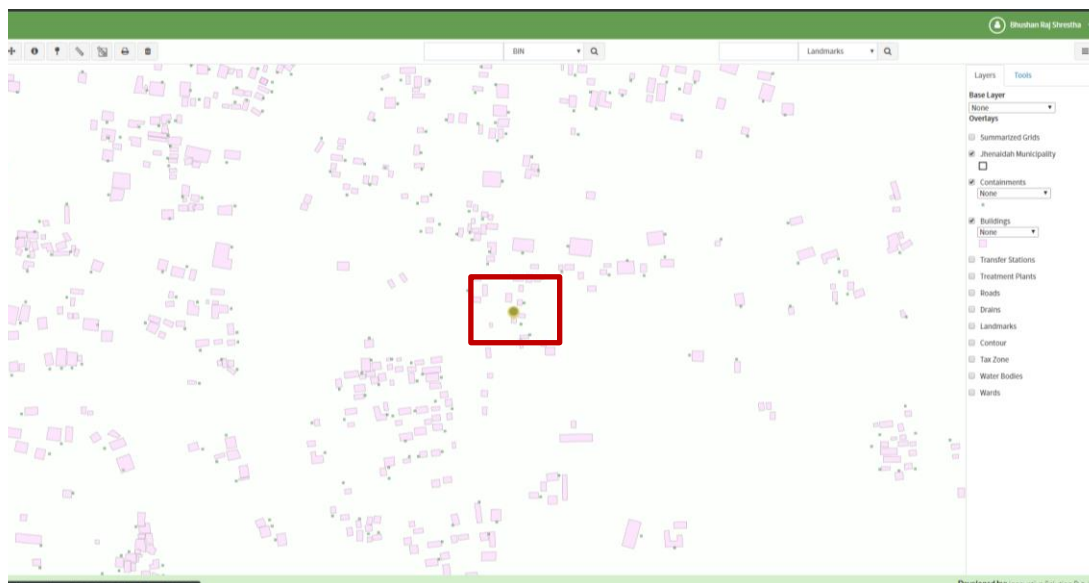
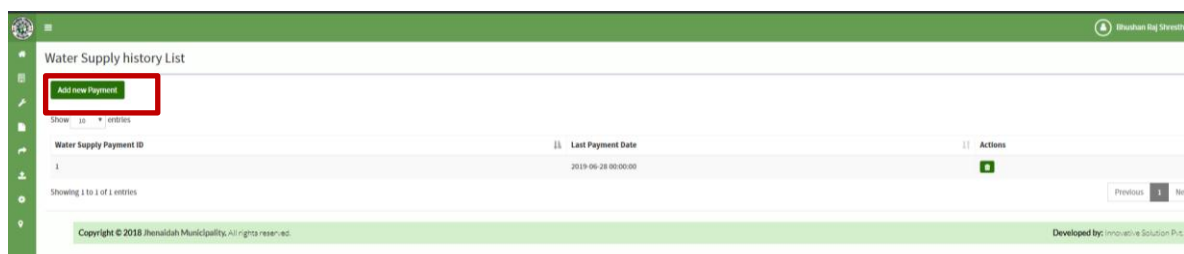


Fig. 9: Building Action: Map

Action: Water Payment

- Click on **Water Payment** button in the Action column of the Building list
- View water payment history of the building in the Water Supply history List page
- Use **Delete** actions to delete a particular record
- Click on **Add new Payment** button in Water Supply history List to add new water payment.
- Fill in a Last Payment Date filed with a date picker



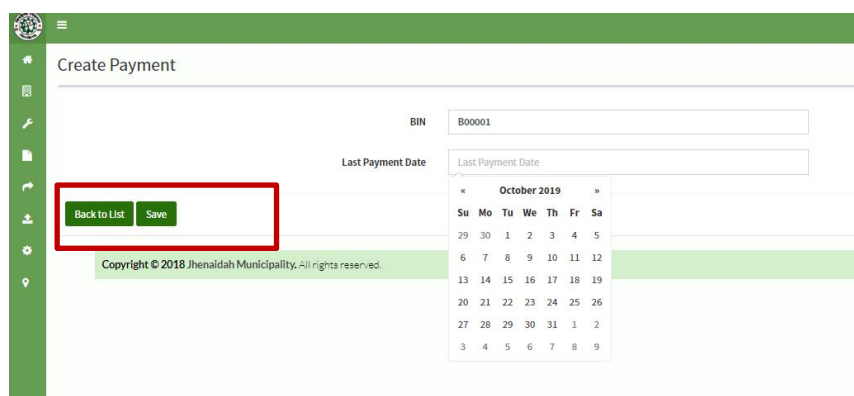


Fig. 10: Building Action: Water Supply Payment

5.1.3 Tools

There are two tools in the Building Structure List page. **Add Building** and **Export to Excel** tool. **Add Building** is used for adding missing or newly constructed building. Missing or newly collected building are collected with **Building and Containment Information Collection System (BCICS) Mobile App** and temporally stored in Building Survey List page (Building Info Management > Building Survey). **Export to Excel** is used to export records in Building Structure List page in an Excel file. It is to be noted that **Export to Excel** works seamlessly with all Filters available in the Building Structure List.

Add Building

- Click on the **Add Building** button
- In the Create Building Structure page, fill in the attributes of the building.
- For KML File, navigate to user's local computer and select an appropriate KML file. Make sure that the selected KML file should contain a single building outline.
 - Building collected with BCICS Mobile App is first stored in Building Survey List page
 - Download the KML in local computer and view in Google Earth
 - Examine/Verify the building outline in Google Earth
 - Make any required modification if any (location/shape) of building outline
 - Save the KML in a new file if there is any modification
 - If there is no modification then use the original KML downloaded
 -
- Click on **Save** button
- Newly added building will be added at the end of list in Building Structure List page

- Examine attributes of the newly added building with **Detail** Action and ensure the building appears on its appropriate location using **Map** Action.

Create Building Structure

Holding Number	
Tax Code	
Road Code	
Ward	
Structure Type	
Floor Count	
Estimated Area of the Building (sqm)	
Functional Use of Building	
Use Categories of Building	
Office or Business Name in Building	
BIN of Main Building	
Surveyed Date	
Draincode	
Drain Connected Date	
Drain Connection	
Water Connection	
Water Supply Customer ID	
Water Connection Date	
Customer	
Contact	
KML File	Choose File No file chosen

Fig. 11: Create Building Structure

Export to Excel

- Filter the records as desired. E.g. Structure Type >Pucca Building, Ward > Ward 1 and Tax Zone> 094.
- Click **Filter** button
- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Building Structure List

Building Identification Number (BIN)
 Tax Zone

Structure Type Pucca
 Functional Use All Functional Use
 Water Connection All Water Connection Status

Ward 9
 Tax Status All Tax Status

Show 10 entries

Building Identification Number (BIN)	Building Number	Road Code	Ward	Structure Type	Floor Count	Functional Use	Tax Code	Tax Zone	Tax Status	Actions
B23890	XXXX-XX	R01561	9	Pucca	1	Residential	09-094-XXXX-XX	094	Dist	Edit Delete
B23891	XXXX-XX	R01569	9	Pucca	1	Educational	09-094-XXXX-XX	094	Dist	Edit Delete
B23892	XXXX-XX	R01565	9	Pucca	1	Educational	09-094-XXXX-XX	094	Dist	Edit Delete
B23893	XXXX-XX	R01561	9	Pucca	1	Residential	09-094-XXXX-XX	094	Dist	Edit Delete
B23894	XXXX-XX	R01563	9	Pucca	1	Educational	09-094-XXXX-XX	094	Dist	Edit Delete
B23895	XXXX-XX	R01558	9	Pucca	1	Educational	09-094-XXXX-XX	094	Dist	Edit Delete
B23896	XXXX-XX	R02255	9	Pucca	1	Educational	09-094-XXXX-XX	094	N/A	Edit Delete
B23897	XXXX-XX	R02255	9	Pucca	1	Educational	09-094-XXXX-XX	094	N/A	Edit Delete
B23898	XXXX-XX	R02255	9	Pucca	1	Residential	09-094-XXXX-XX	094	Dist	Edit Delete
B23899	XXXX-XX	R01558	9	Pucca	1	Residential	09-094-XXXX-XX	094	N/A	Edit Delete

Showing 1 to 10 of 22 entries (Filtered from 25,117 total entries)

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	A	B	C	D	E	F	G	H	I	J	K	L	Q	f
1	bin	holding	taxcd	roadcd	ward	structype	flrcount	bldgarea	bldguse	usecatg	offcnm	bldgasc		
2	B23890	XXXX-XX	09-094-XX	R01561	9	Pucca	1	54.1658	Residential					
3	B23891	XXXX-XX	09-094-XX	R01569	9	Pucca	1	253.722	Educational					
4	B23892	XXXX-XX	09-094-XX	R01565	9	Pucca	1	283.048	Educational					
5	B23893	XXXX-XX	09-094-XX	R01561	9	Pucca	1	53.8881	Residential					
6	B23894	XXXX-XX	09-094-XX	R01563	9	Pucca	1	130.826	Educational					
7	B23895	XXXX-XX	09-094-XX	R01558	9	Pucca	1	157.097	Educational					
8	B23896	XXXX-XX	09-094-XX	R02255	9	Pucca	1	486.797	Educational					
9	B23897	XXXX-XX	09-094-XX	R02255	9	Pucca	1	363.111	Educational					
10	B23898	XXXX-XX	09-094-XX	R02255	9	Pucca	1	1486.61	Residential					
11	B23899	XXXX-XX	09-094-XX	R01558	9	Pucca	1	29.076	Residential					
12	B23900	XXXX-XX	09-094-XX	R01558	9	Pucca	1	39.6021	Educational					
13	B23901	XXXX-XX	09-094-XX	R01558	9	Pucca	1	33.9579	Educational					

Fig. 12: Export to Excel

5.2 Building Surveys

- Navigate to Building Info Management > Building Surveys

Maintains database of missing or newly collected buildings that are collected with **BCICS Mobile App**.

Building Survey List

Show 10 entries

search:

Building Identification Number	Tax Code	Survey Date	Actions
B90001		2018-11-05	Edit Delete
B90003		2018-11-05	Edit Delete
B90002		2018-11-05	Edit Delete
B90004		2018-11-05	Edit Delete
B90005		2018-11-05	Edit Delete
99991		2018-11-14	Edit Delete
99991		2018-11-14	Edit Delete
5555		2018-11-18	Edit Delete
B90006		2018-11-18	Edit Delete
B0007		2018-11-19	Edit Delete

Showing 1 to 10 of 19 entries

Previous 1 2 Next

Fig. 13: Building Survey List

5.2.1 Actions

ICON	Name	Task that will be performed
	Download	Downloads the KML file of the building
	Delete	Deletes the Building Survey Record

6 Utility Information Management

This is a module dedicated to support maintaining database of utilities such as road and drainage network.

6.1 Roads

- Navigate to Utility Info Management > Roads

Road List

Road Code	Road Name	Road Hierarchy	Actions
R00001		Local Road	[Icons]
R00002	Shekherpur Road	Major Road	[Icons]
R00003		Local Road	[Icons]
R00004		Local Road	[Icons]
R00005		Local Road	[Icons]
R00006		Local Road	[Icons]
R00007		Local Road	[Icons]
R00008	Shekherpur Road	Major Road	[Icons]
R00009		Local Road	[Icons]
R00010		Local Road	[Icons]

Showing 1 to 10 of 2,312 entries

Previous 1 2 3 4 5 ... 232 Next

Fig. 14: Road List

Only some of the important attributes are shown in the list. Attributes shown can be sorted in ascending/descending order. The user can search for a particular road according to Road Code. There are filters available that enables user to Road List according to their need. There are different Actions buttons available for each record.

6.1.1 Filter

- Enter a specific Road Code in textbox Click on Filter button. To remove filter, click Reset.
- Choose Road Hierarchy Filter, select one option (All Road, Major Road or Local Road) and click on Filter button. To remove filter, click Reset.

Export to Excel

Road Code Road Hierarchy

Show entries

Road Code	Road Name	Road Hierarchy	Actions
R00007		Local Road	   

Showing 1 to 1 of 1 entries (filtered from 2,312 total entries) Previous

Road List

Export to Excel

Road Code Road Hierarchy

Show entries

Road Code	Road Name	Road Hierarchy	Actions
R00001		Local Road	   
R00003		Local Road	   
R00004		Local Road	   
R00005		Local Road	   
R00006		Local Road	   
R00007		Local Road	   
R00009		Local Road	   
R00010		Local Road	   
R00011		Local Road	   
R00012		Local Road	   

Showing 1 to 10 of 2,058 entries (filtered from 2,312 total entries) Previous 1 2 3 4

Fig. 15: Using Filter to view specific road according to its Road Code and Road Hierarchy

6.1.2 Actions

There are four different Actions available. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there is a need of modifying any particular attribute.

Table 2: Actions for road data

ICON	Name	Task that will be performed
	Edit	Opens form to edit the road details.
	Detail	Shows detail information about the road
	Delete	Deletes road data
	Map	Shows spatial location of the road on a map

Action: Edit

- Click on the **Edit** button in the Action column of the Road list
- In Edit Road page, enter empty fields or change/modify already added fields as necessary
- Click on **Save** button if the user with to save any changes. After saving, the user will be directed to Road List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Road List page

Edit Road

Road Code	R00001
Road Name	Road Name
Road Hierarchy	Local Road
Road Surface Type	
Road Length (in meter)	700.36
Road Width (in meter)	7
Carrying Width of the Road (in meter)	Carrying Width of the Road (in meter)

Fig. 16: Road Action- Edit

Action: Detail

- Click on the **Detail** button in the Action column of the Road List page
- Explore attributes of a road as desired
- Click on **Back to List** to go to Road List page

Road Details

Back to List

Road Code	R00001
Road Name	
Road Hierarchy	Local Road
Road Surface Type	
Road Length (in meter)	700.36
Road Width (in meter)	
Carrying Width of the Road (in meter)	

Fig. 17: Road Action-Detail

Actions: Delete

- Click on the **Delete** button in the Action column of the Road List table.
- It will display confirmation dialog
- Click OK button in the confirmation dialog.

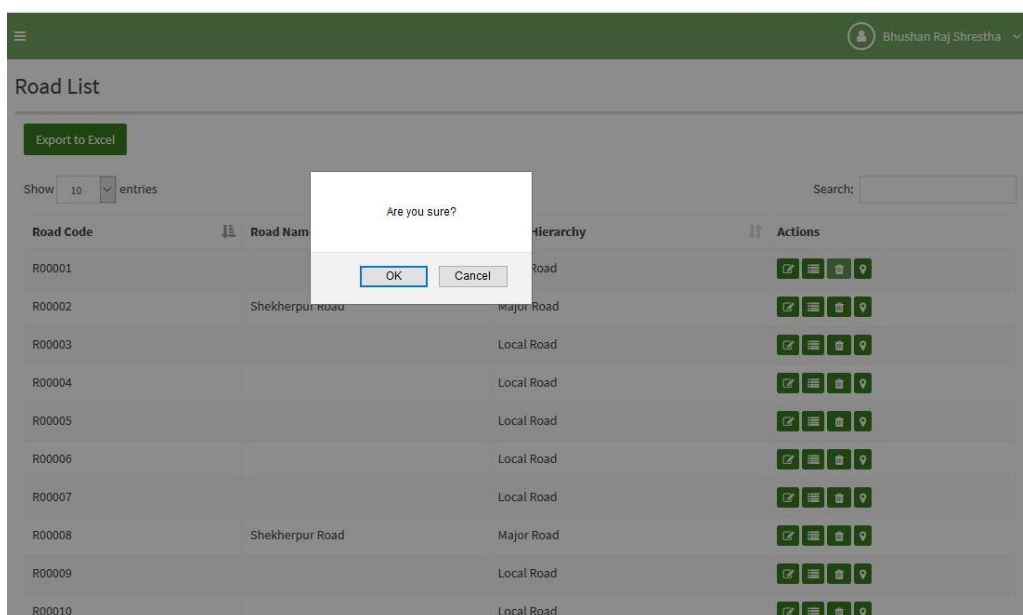


Fig. 18: Road Action-Delete

Action: Map

- Click on **Map** button in the Action column of the Road List. A map will open, showing a spatial location of the road highlighted in red color
- To go back to Road List page, Navigate to Utility Info Management > Roads

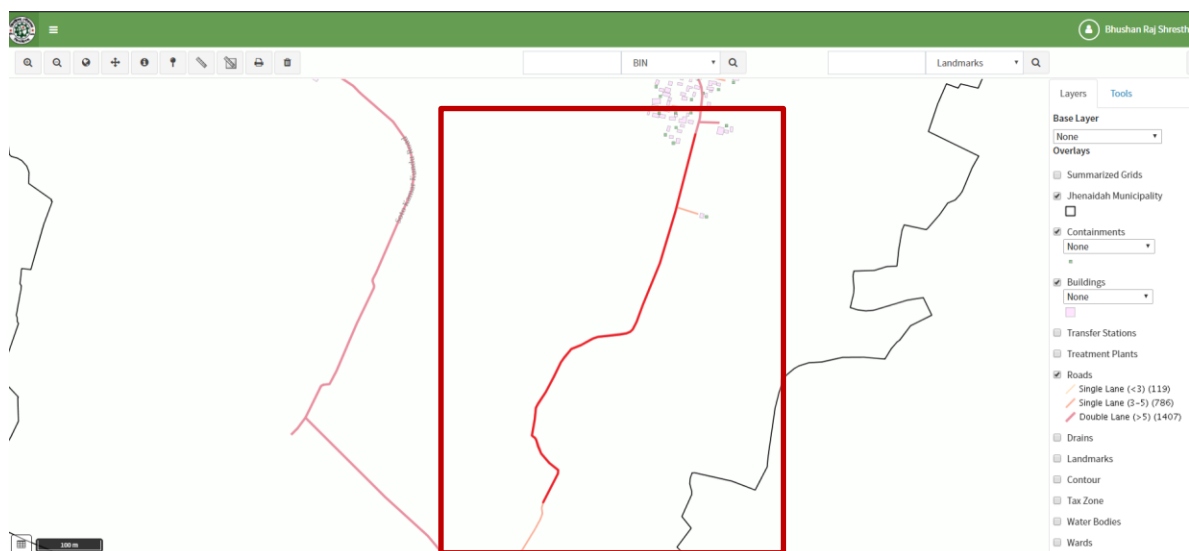


Fig. 19: Road Action-Map

6.1.3 Tools

Export to Excel is used to export records in Road List page in an Excel file. It is to be noted that **Export to Excel** works seamlessly with Filter available in the Road List page.

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

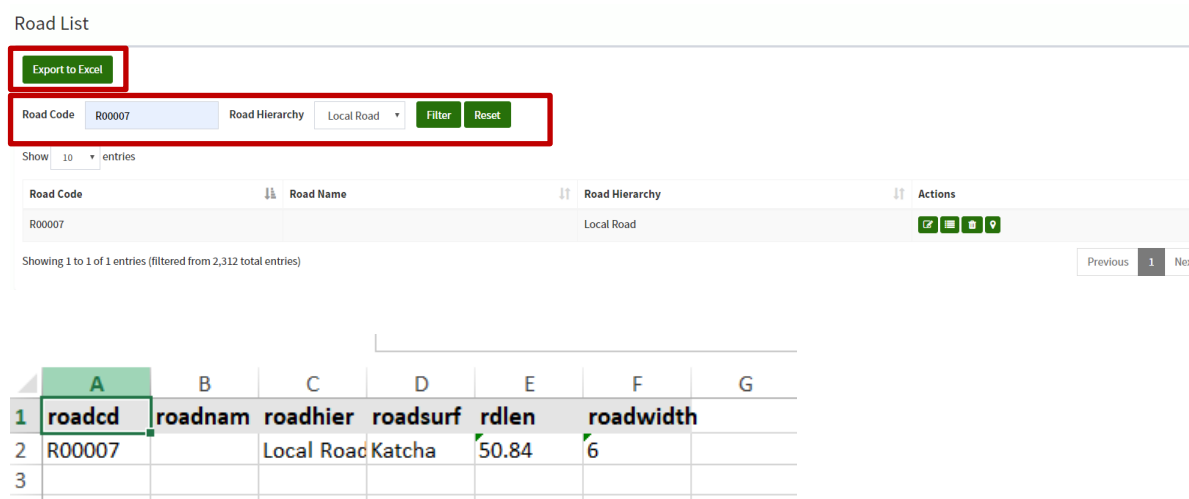


Fig. 20: Road - Export to Excel tool

6.2 Drains

- Navigate to Utility Info Management > Drains

Drain List

Tools

Export to Excel

Drain Code Drain Type All Types **Filter** **Reset**

Filter

Show 10 entries

Drain Code	Drain Type	Actions
D00001	Open	   
D00002	Closed	   
D00003	Open	   
D00004	Closed	   
D00005	Open	   
D00006	Closed	   
D00007	Closed	   
D00008	Closed	   
D00009	Closed	   
D00010	Open	   

Showing 1 to 10 of 273 entries

Previous 1 2 3 4

Actions

Fig. 21: Drain List

Only some of the important attributes are shown in the list. Attributes shown can be sorted in ascending/descending order. The user can search for a particular drain according to Drain Code. There are filters available that enables user to Drain List according to their need. There are different Actions buttons available for each record.

6.2.1 Filter

- Enter a specific Drain Code in textbox Click on Filter button. To remove filter, click Reset.
- Choose Drain Type Filter, select one option (All Types, Open, Closed or Unknown) and click on Filter button. To remove filter, click Reset.

Drain List

Export to Excel

Drain Code Drain Type All Types **Filter** **Reset**

Show 10 entries

Drain Code	Drain Type	Actions
D00010	Open	   

Showing 1 to 1 of 1 entries (filtered from 273 total entries)

Drain List

[Export to Excel](#)

Drain Code Drain Type [Filter](#) [Reset](#)

Show entries

Drain Code	Drain Type	Actions
D00001	Open	Edit Detail Delete Map
D00003	Open	Edit Detail Delete Map
D00005	Open	Edit Detail Delete Map
D00010	Open	Edit Detail Delete Map
D00013	Open	Edit Detail Delete Map
D00014	Open	Edit Detail Delete Map
D00018	Open	Edit Detail Delete Map
D00022	Open	Edit Detail Delete Map
D00027	Open	Edit Detail Delete Map
D00032	Open	Edit Detail Delete Map

Showing 1 to 10 of 178 entries (filtered from 273 total entries)

[Previous](#) [1](#) [2](#)

Fig. 22: Using Filter to view specific drain according to its Drain Code and Drain Type

6.2.2 Actions

There are four different Actions available for building. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there a need of modifying any particular attribute.

Table 3: Actions for Drain data

ICON	Name	Task that will be performed
	Edit	Opens form to edit the drain details.
	Detail	Shows detail information about the drain
	Delete	Deletes drain data
	Map	Shows spatial location of the drain on a map

Action: Edit

- Click on the **Edit** button in the Action column of the Drain list
- In Edit Drain page, enter empty fields or change/modify already added fields as necessary

- Click on **Save** button if the user with to save any changes. After saving, the user will be directed to Road List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Road List page

Edit Drain

Drain Code	D00001
Drain Category	
Drain Type	Open
Capacity (cum)	Capacity (cum)
Length (m)	Length (m)
Lining	
Canal Name	Canal Name
Canal Type	

Fig. 23: Drain Action- Edit

Action: Detail

- Click on the **Detail** button in the Action column of the Drain List page
- Explore attributes of a drain as desired
- Click on **Back to List** to go to Road List page

Drain Details

Drain Code	D00001
Drain Category	
Drain Type	Open
Capacity (cum)	
Length (m)	
Lining	
Canal Name	
Canal Type	

Fig. 24: Drain Action-Detail

Action: Delete

- Click on the **Delete** button in the Action column of the Drain List table.
- It will display confirmation dialog
- Click OK button in the confirmation dialog.

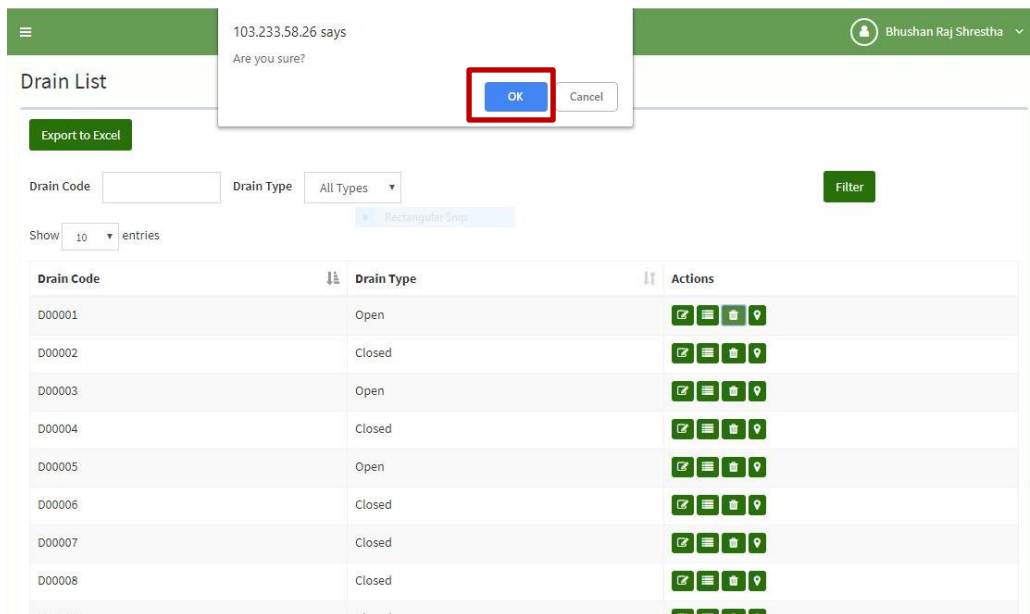


Fig. 25: Drain Action-Delete

Action: Map

- Click on **Map** button in the Action column of the Drain List. A map will open, showing a spatial location of the drain highlighted in red color
- To go back to Road List page, Navigate to Utility Info Management > Roads

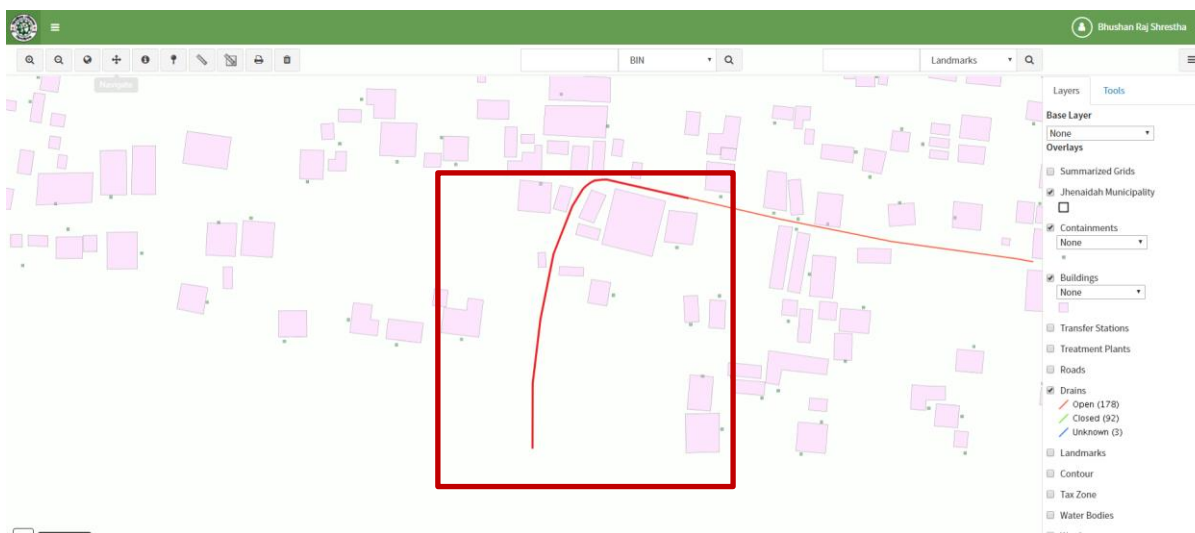


Fig. 26: Drain Action-Map

6.2.3 Tools

Export to Excel is used to export records in Drain List page in an Excel file. It is to be noted that **Export to Excel** works seamlessly with Filter available in the Drain List page.

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Drain List

Export to Excel

Drain Code: D00009 Drain Type: All Types **Filter** **Reset**

Show 10 entries

Drain Code	Drain Type	Actions
D00009	Closed	   

A	B	C	D	E	F	G
draincode	drainsize	drainlen	draincode	draincons		
Closed	4	246.05	D00009	RCC		

Fig. 27: Drain - Export to Excel tool

7 FSM Information Management

This is a module is dedicated to support a complete cycle for an efficient FSM.

7.1 Containments

- Navigate to FSM Info Management > Containments

7.1.1 Filters

- Enter a specific Containment Code or Road Code and click on **Filter** button.
- Choose an appropriate Filter as desired, select its appropriate value and click on **Filter** button. E.g. Containment Type > Septic Tank and Ward > Ward 1. To remove filter, click **Reset**.

Containment Code:

Containment Size:

Containment Type:

Customer Name:

Road Code:

Ward:

Filter

Containment Code	Containment Type	Size	Road Code	Road Name	Carrying Width	Customer Name	Ward	Actions
C02929	Septic Tank	0	R01859		0		1	
C02946	Septic Tank	0	R01137	Keramot ALI Sorok	0		1	
C02958	Septic Tank	0	R01860		0		1	
C02959	Septic Tank	0	R01137	Keramot ALI Sorok	0		1	
C02962	Septic Tank	0	R01137	Keramot ALI Sorok	0		1	

Fig. 28: Using Filter to view specific containments

7.1.2 Actions

ICON	Name	Task that will be performed
	Building	It will display the list of buildings connected to the containment
	Edit	Opens form to edit the containment details
	Details	Shows detail information about the containment
	History	Shows information about log history of edits
	Delete	Delete containment data
	Map	Shows spatial location of the containment on a map

	Emptying Service History	Shows Emptying Service history of the containment
	Create Application	Creates an application for the containment
	Nearest Road	Show nearest road to the containment in a map

Building

This action shows a **Building Connected to Containment** page that shows a list of buildings that are connected to a containment. If a single building is connected to a containment, connected building automatically assigned as main building. A containment may serve more than one building so in that case there can be multiple buildings in the list. In that case, a building has to be set by **Make Main** Action.

Building Connected to Containment: C00001

Back to the Containment List Add Buildings to this Containment View in Map						
Building Identification Number (BIN)	Holding Number	Structure Type	Estimated Area of the Building	BIN of Main Building	Is Main Building	Actions
B19665	XXXX-XX	Katcha	61.2071		✓	Make Main 

Fig. 29: Containment Action - Building (A single building connected to a containment)

Building Connected to Containment: C00004

Buildings added to this containment						
Back to the Containment List Add Buildings to this Containment View in Map						
Building Identification Number (BIN)	Holding Number	Structure Type	Estimated Area of the Building	BIN of Main Building	Is Main Building	Actions
B00017	0164-00	Semipucca	47.7896			Make Main 
B00018	XXXX-XX	Semipucca	44.7061			Make Main 
B00019	XXXX-XX	Semipucca	47.0519			Make Main 

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Fig. 30: Containment Action - Building (Multiple buildings connected to a containment)

- Click on the **Building** button in the Action column of the Containment List
- A list of buildings connected to the containment is shown in Building Connected to Containment page. If a building is needed to be disconnected with the containment, click **Delete** Action on that page. If a building is needed to be connected with the

containment, click **Add Building to this containment** Action. A new Add Buildings to Containment page will be displayed. Select the building to be added with **ADD** Action and click **Submit** button.

- Click on the **View in Map** button in Building Connected to Containment page to view connected building in a map

Add Buildings to Containment: C00002

[Back to the added Buildings list](#)

Selected Buildings
[Submit](#)

Search to Select Buildings
 Building Identification Number (BIN) [Filter](#)
 Show entries Search:

Building Identification Number (BIN)	Holding Number	Structure Type	Estimated Area of the Building	BIN of Main Building	Actions
B00001	XXXX-XX	Katcha	30.1617		Add
B00002	XXXX-XX	Katcha	28.6534		Add
B00003	XXXX-XX	Katcha	42.8961		Add
B00004	XXXX-XX	Katcha	31.837		Add
B00005	0208-00	Semipucca	54.2911		Add

Fig. 31: Adding building to a containment

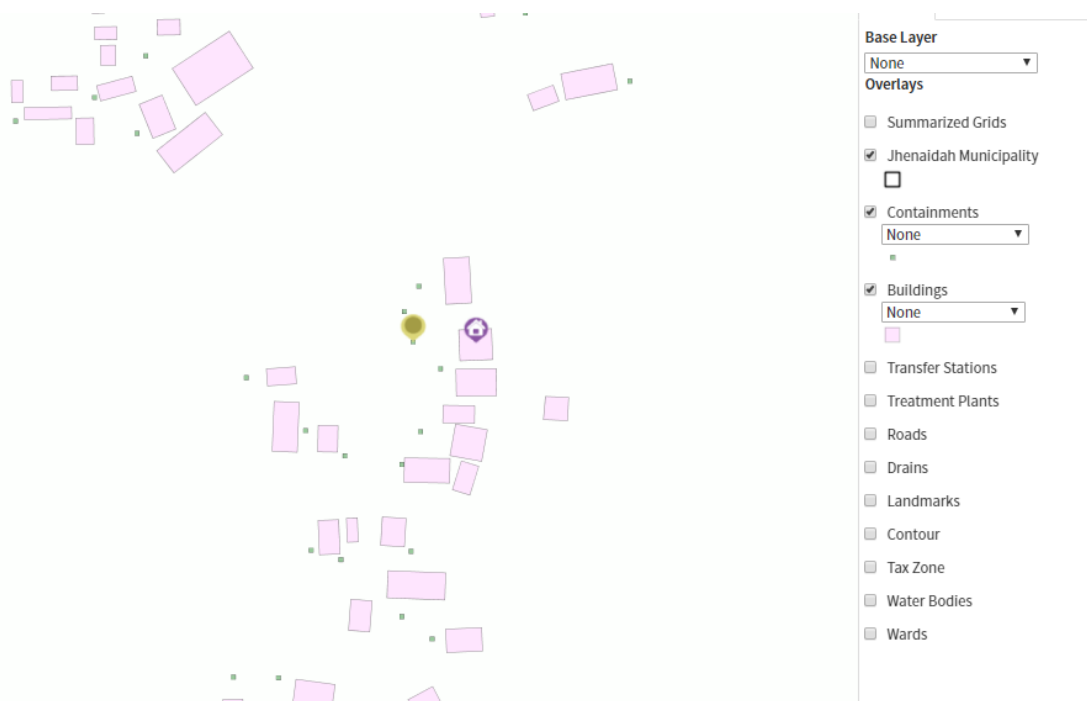


Fig. 32: Showing building connected to a containment

Edit

- Click on the **Edit** button in the Action column of the Containment List
- In Edit Containment List page, enter empty fields or change/modify already added fields as necessary
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Containment List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Containment List page

Edit Containment

Containment Code: C00001

Containment Type	Pit ▼
Containment Size (cum)	Containment Size (cum)
Ward	7 ▼
Buildings Served	1
Household Served	2
Population Served	25
Road Code	R00592
Road Name	Mondol Para Road
Road Width (m)	10
Carrying Width of the Road (m)	Carrying Width of the Road (m)
Vacutug Type	Vacutug Type
Transfer Station Code	Transfer Station Code

[Back to List](#) | [Save](#)

Fig. 33: Containments Action - Edit

Detail

- Click on the **Detail** button in the Action column of the Containment List
- Explore attributes of Containment record as desired
- Click on **Back to List** button to go to Containment List

History

- Click on the **History** button in the Action column of the Containment List
- View the changes made to attributes of the containment
- Click on **Back to List** to go to Containment List page

Containment History

Back to List
• Bhushan Raj Shrestha changed popsrv from 10 to 25 on 2019-07-27 10:07:30 • Bhushan Raj Shrestha changed customer from to Bhushan Raj Shrestha on 2019-07-16 08:55:46

Fig. 34: Containment Action: History

Delete

- Click on the **Delete** button in the Action column of the containment List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

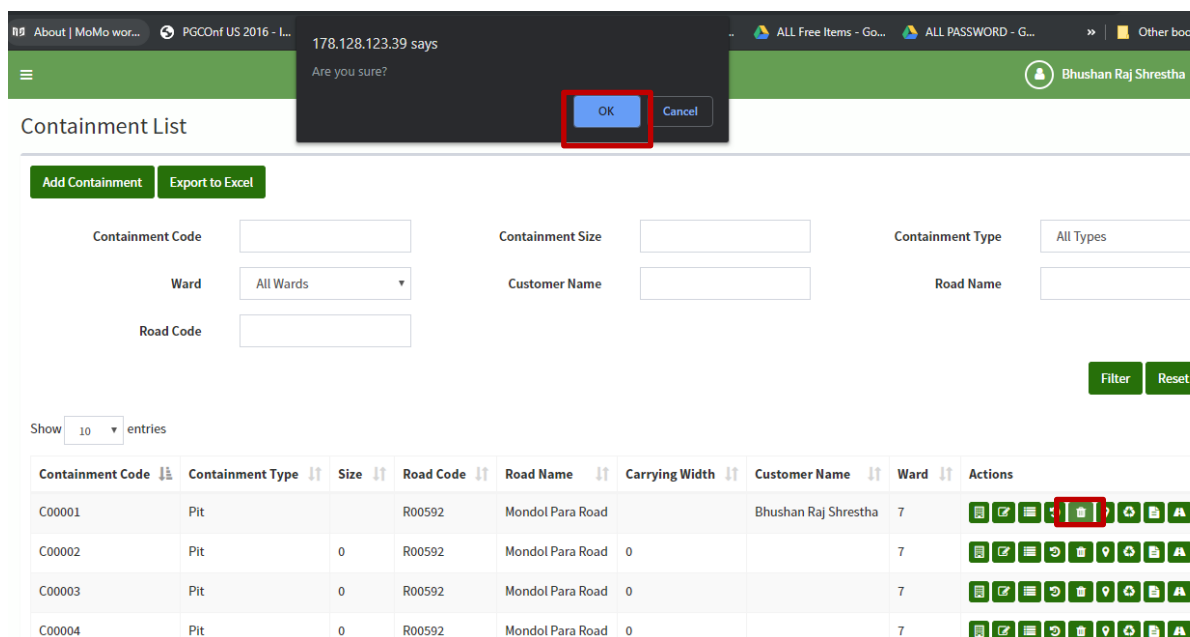


Fig. 35: Containments Action - Delete

Map

- Click on **Map** button in the Action column of the Containment List. A map will open, showing a spatial location of the Containment with a marker
- To go back to Containment List page, either click Back button on the browser or navigate to FSM Info Management > Containments.

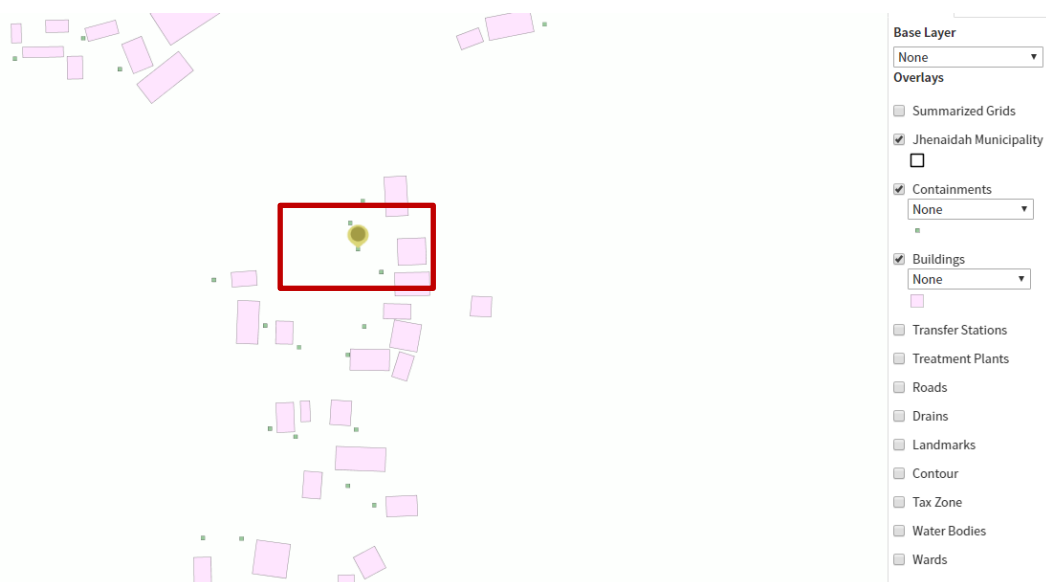


Fig. 36: Containments Action: Map

Emptying Service History

- Click on the **Emptying Service History** button in the Action column of the Containment List
- View the Emptying Service History of the containment. There is a Detail Action that can be clicked to view Emptying Service History detail of that containment.

Emptying Service History

Show

10

entries

Search:

Containment Code	Emptying Date	Start Time	End Time	Uploaded By	Actions
C10000	2018-11-18	02:38:06	05:38:09	sichu	
C10000	2018-11-18	03:36:46	04:36:49	sichu	

Fig. 37: Containment Action: Emptying Service History

Create Application

Some of the attribute of containment may be empty. So, it's a good practice to fill details of containments when an application for emptying request is received. Once an application is saved, the newly created application will be appended at the end of Application List (FSM Info Management > Applications).

- Click on the **Create Application** button in the Action column of the Containment List
- A Create Application page will open. Fill in empty fields. Please note that all fields in Create Application field are mandatory.

- Click **Save** to create an application.

Create Application

Containment Code	C00001
Containment Type	Pit
Application Date	Application Date (yyyy-mm-dd)
Holding Number	Holding Number
Customer Name	Customer Name
Father/Mother Name	Father/Mother Name
Husband/Wife Name	Husband/Wife Name
Ward	7
Road	Road
Address	Address
Mobile	Mobile
Number of Pit	Pit Number
Pit Diameter/Width	Pit Diameter
Pit Depth	Pit Depth

Back to List Save

Fig. 38: Containment Action - Create Application

Nearest Road

- Click on the Nearest button in the Action column of the Containment List
- A map will open that will show the nearest road to the containment. Use **Info** tool to get information about the Nearest Road, if needed. Use **Measure Distance** tool to get distance information between the containments and its corresponding nearest road, if needed.
- To go back to Containment List page, either click Back button on the browser or navigate to FSM Info Management > Containments.

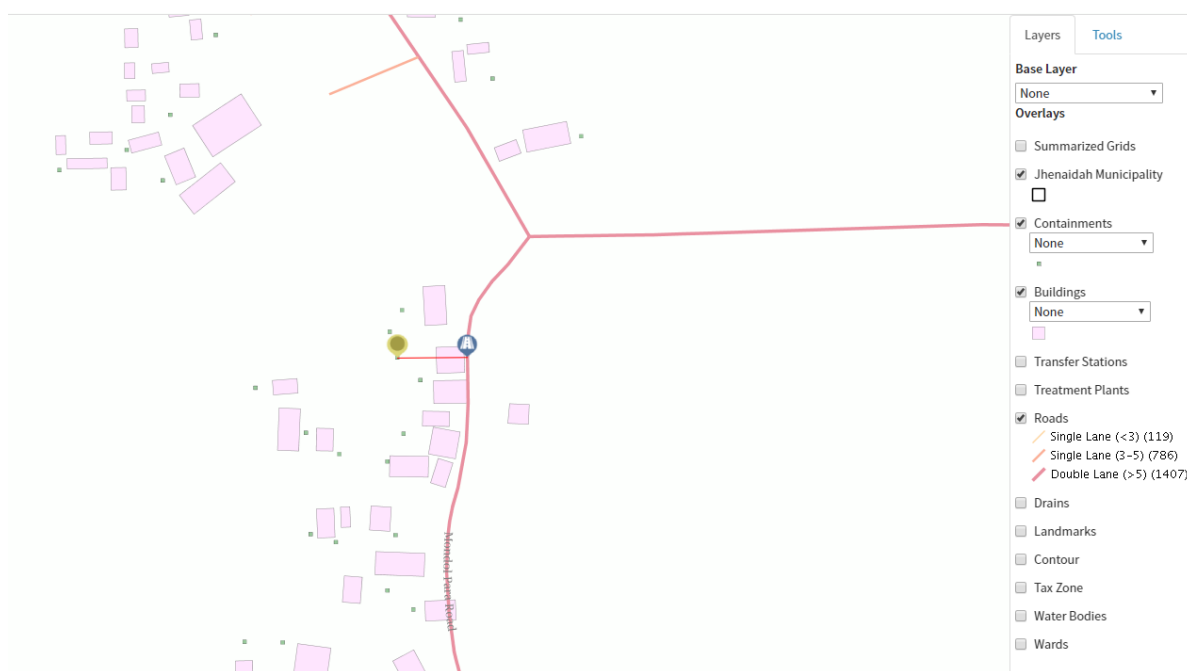


Fig. 39: Containment Actions: Nearest Road

7.1.3 Tools

Add Containment

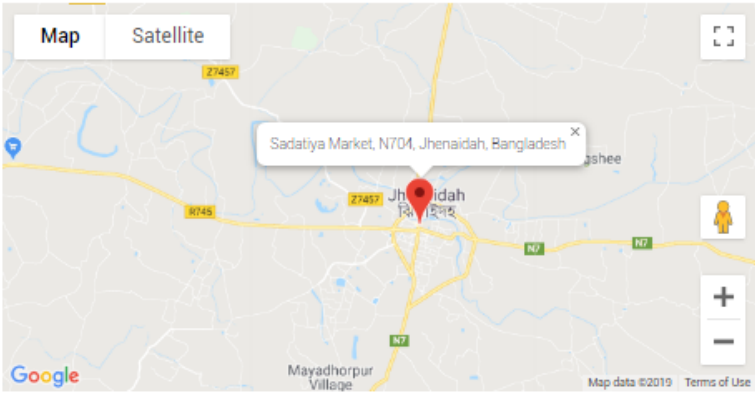
This tool is used for adding new/missing containment in IMIS if the user knows well an exact location of the containment. In that case, the user can set location of the new/missing containment in a map by dragging markers without actually visiting the field and without using BCICS. But it is recommended that the user actually visit the field and use BCICS app for containment data collection.

- Click on **Add Containment** button.
- Then fill the required data. Set a location of the containment by dragging a red marker on the map. Notice that Longitude and Latitude fields are automatically filled as the user drag red marker on the bottom map. Click **Save** button.
- Newly added Containment record will be added at the end of the list

Create Containment

Containment Type	
Containment Size (cum)	Containment Size (cum)
Ward	
Buildings Served	Buildings Served
Household Served	Household Served
Population Served	Population Served
Road Code	Road Code
Road Name	Road Name
Road Width (m)	Road Width (In meter)
Carrying Width of the Road (m)	Carrying Width of the Road (m)
Vacutug Type	Vacutug Type
Transfer Station Code	Transfer Station Code
Longitude	89.17362000000003
Latitude	23.54676

Drag Marker To Set Latitude And Longitude



Back to List Save

Fig. 40: Containment Tool - Add Containment

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Containment List

Containment Code:
 Containment Size:
 Containment Type:

Ward:
 Customer Name:
 Road Name:

Road Code:

Show entries

Containment Code	Containment Type	Size	Road Code	Road Name	Carrying Width	Customer Name	Ward	Actions
C00001	Pit		R00592	Mondol Para Road		Bhushan Raj Shrestha	7	   
C00002	Pit	0	R00592	Mondol Para Road	0		7	   
C00003	Pit	0	R00592	Mondol Para Road	0		7	   
C00004	Pit	0	R00592	Mondol Para Road	0		7	   
C00005	Pit	0	R00592	Mondol Para Road	0		7	   

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	
1	containcd	containtp	containsz	ward	buildsrv	hhsrv	popsrv	roadcd	roadnm	rowwidth	rdcarwidth	vacutug	transstncc	customer	contact
2	C00001	Pit		7	1	2	25	R00592	Mondol Pz	10				Bhushan Raj Shresth	
3	C00002	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
4	C00003	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
5	C00004	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
6	C00005	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
7	C00006	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
8	C00007	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
9	C00008	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
10	C00012	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
11	C00013	Pit	0	7		0	0	R00592	Mondol Pz	10	0				
12	C00014	Pit	0	7		0	0	R00592	Mondol Pz	10	0				

Fig. 41: Containments Tool - Export to Excel

7.2 Containment Surveys

Maintains database of missing or newly collected containments that are collected with **BCICS Mobile App**.

- Navigate to FSM Info Management > Containment Surveys

Containment Survey List

Show entries

Search:

Containment Code	Longitude	Latitude	Survey Date	Actions
C0007	89.21277835965157	23.166153088087043	2018-11-19	   
C101	85.311442	27.7153048	2019-09-30	  
C80801	89.17671203613281	23.510478175601378	2018-11-05	  
C80802	89.18134689331055	23.52842183815669	2018-11-05	  
C80803	89.18838500976564	23.551398745203414	2018-11-05	  
C80804	89.15405273437499	23.551084020172624	2018-11-05	  
C80805	89.1701889038086	23.56681934847647	2018-11-05	  

Fig. 42: Containment Survey List

7.2.1 Filters

- Type the search term in the search filter. The list will be dynamically updated to match search term
- Delete search term to get back to the previous list

Show entries

Search:

Containment Code	Longitude	Latitude	Survey Date	Actions
C101	85.311442	27.7153048	2019-09-30	  

Fig. 43: Using search filter to find a surveyed containment on particular year

7.2.2 Actions

ICON	Name	Task that will be performed
	Approve	Approves the surveyed containment
	Delete	Deletes the surveyed containment record
	Map	Shows spatial location the surveyed containment on a map

Approve

- Click on the **Approve** button in the Action column of the Containment Survey List.
- It will display form asking for containment details. Fill the required details and click **Save** button. The approved containment will be appended at end of Containment List (FSM Info Management > Containment). Use Building Action in Containment List to assigned a building to that particular containment.

Add Containment

Containment Type	
Containment Size (cum)	
Ward	
Buildings Served	
Household Served	
Population Served	
Road Code	
Road Name	
Road Width (m)	
Carrying Width of the Road (m)	
Vacutug Type	
Transfer Station Code	
Longitude	89.1701889038086
Latitude	23.56681934847647

Fig. 44: Containment Survey Action - Approve

Delete

- Click on the **Delete** button in the Action column of the Containment Survey List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

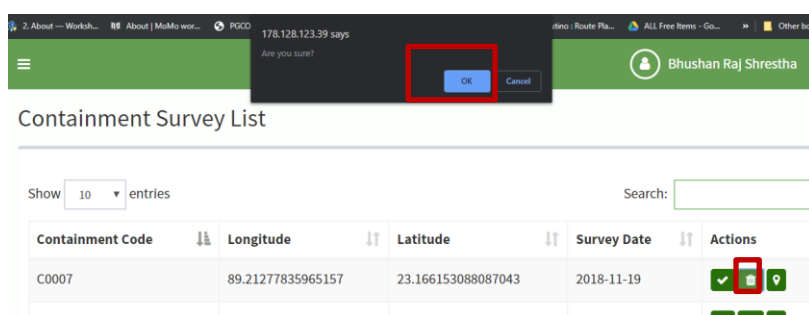


Fig. 45: Containment Survey Action - Delete

Map

- Click on **Map** button in the Action column of the Containment Survey List. A map will open, showing a spatial location of the surveyed containment with a marker
- To go back to Containment page, either click Back button on the browser or navigate to FSM Info Management > Containment Survey.



Base Layer
None

Overlays

☐ Summarized Grids

☒ Jhenaidah Municipality

☐ Containments
None

☒ Buildings
None

☐ Transfer Stations

☐ Treatment Plants

☐ Roads

Fig. 46: Containment Survey Action: Map

7.3 Applications

Maintains database of applications received for containment emptying request. The application request is entered through Create Application Action in FSM Info Management > Containment Section.

- Navigate to FSM Info Management > Applications

Application List

Export to Excel

Tools

Assessment detail Completed All Assessment details Emptying detail Completed All Emptying detail Feedback detail Completed All Feedback detail

Ward All Wards

Filter

Filter

Reset

Actions

Application ID	Containment Code	Customer Name	Road	Ward	Assessment detail Completed	Emptying detail Completed	Feedback detail Completed	Actions
11	C00045			7	✓	✓	✓	[Icons]
12	C00010	Ishlam Mehaboob		7	✓	✓	✓	[Icons]
13	C00008			7	✓	✓	✗	[Icons]
14	C00042			7	✓	✓	✗	[Icons]
16	C00049			7	✓	✗	✗	[Icons]
17	C00091			7	✓	✓	✗	[Icons]
19	C00097			7	✓	✗	✗	[Icons]
21	C00099			7	✓	✗	✗	[Icons]

Fig. 47: Application List for containment emptying request

7.3.1 Filters

- Choose an appropriate Filter as desired, select its appropriate value and click on **Filter** button. E.g. Assessment detail Completed = YES, Ward = Ward 7. To remove filter, click **Reset**.

Assessment detail Completed: Yes
Ward: 7

Emptying detail Completed: All Emptying detail
Feedback detail Completed: All Feedback detail

Filter Reset

Show 10 entries

Application ID	Containment Code	Customer Name	Road	Ward	Assessment detail Completed	Emptying detail Completed	Feedback detail Completed	Actions
11	C00045			7	✓	✓	✓	[Icons]
12	C00010	Ishlam Mehaboob		7	✓	✓	✓	[Icons]
13	C00008			7	✓	✓	✗	[Icons]

Fig. 48: Using search filter to find specific application status and ward

7.3.2 Actions

There are seven different Actions available for each record. Many containment details are not complete. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there a need of modifying any particular attribute. Assessment details get populated with BSCICS app. Once an assessment details for an application is populated, Emptying Service Details Action gets enabled. After filling up Emptying Service detail, Feedback Detail gets enabled. Both Filled Emptying Service Details & Feedback Details forms are received as a hard copy and later entered into IMIS.

Table 4: Actions for Applications

ICON	Name	Task that will be performed
	Edit	Opens form to edit the Containment record details.
	Detail	Shows detail information about Containment record
	Delete	Deletes an Application record
	Map	Shows spatial location of the containment on a map
	Assessment Details	Shows Assessment details
	Emptying Service Details	Fill & show Emptying Service details
	Feedback Details	Fill & show Feedback Details

Edit

- Click on the **Edit** button in the Action column of the Application List

- In Edit Application page, enter empty fields or change/modify already added fields as necessary. Notice that information about Pit/Septic tank details are collected at stage and Edit Application page changes according to Containment Type (Pit/ Septic Tank)
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Application List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Application page

Edit Application

Containment Code	C00010
Containment Type	Pit
Application Date	Application Date (yyyy-mm-dd)
Holding Number	Holding Number
Customer Name	Customer Name
Father/Mother Name	Father/Mother Name
Husband/Wife Name	Husband/Wife Name
Ward	7
Road	Road
Address	Address
Mobile	Mobile
Number of Pit	Pit Number
Pit Diameter/Width	Pit Diameter
Pit Depth	Pit Depth

[Back to List](#) [Save](#)

Fig. 49: Application Action – Edit for Containment Type Pit

Edit Application

Containment Code	C00476
Containment Type	Septic Tank
Application Date	Application Date (yyyy-mm-dd)
Holding Number	Holding Number
Customer Name	Customer Name
Father/Mother Name	Father/Mother Name
Husband/Wife Name	Husband/Wife Name
Ward	7
Road	Road
Address	Address
Mobile	Mobile
Tank Length	Tank Length
Tank Width	Tank Width
Tank Depth	Tank Depth

[Back to List](#)
[Save](#)

Fig. 50: Application Action – Edit for Containment Type Septic Tank

Detail

- Click on the **Detail** button in the Action column of the Application List
- Explore attributes of Containment as desired
- Click on **Back to List** button to go to Application List

Delete

- Click on the **Delete** button in the Action column of the Application List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

Map

- Click on **Map** button in the Action column of the Application List. A map will open, showing a spatial location of the Containment with a marker
- To go back to Application List page, either click Back button on the browser or navigate to FSM Info Management > Applications

Assessment Details

- Click on the **Assessment Detail** button in the Action column of the Application List
- Click on **Back to List** button to go to Application List

Assessment Details

[Back to List](#)

Application ID	10
Service Provider	
Estimated Cost	
Distance From Road (ft)	20
Road Width (ft)	3.5
Assessed Sludge Volume(cum)	34
Vacutug Accessibility?	No
Size of Recommended Vacutug	
Required Trips	3
Comments	It was good
Last Emptied Date	
Proposed Emptying Date	
Uploaded By	sagar

Fig. 51: Application Action – Assessment Details

Emptying Service Details

- Click on the **Emptying Service Detail** button in the Action column of the Application List
- Fill the necessary details and click **Save** button
- Click Back to List button if the user wants to go back to Application List without saving

Emptying Service Details

Application Id	17
Containment Code	C00091
Emptied Date	2019-09-18
Next Emptying Date	2019-09-03
Vacutug No	
Capacity	12
Driver	2
Emptier 1	1
Emptier 2	4
Volume of Sludge (cum)	14
Start Time	12:19 PM
End Time	12:19 PM
Did you use Mobile Transfer Station (MTS)?	
Time Required for Emptying Service After Arrival of Vacutug (in hr)	
Required Trip for Total Emptying	
Receipt Number	
Cost Paid by Containment Owner with Receipt	
Cost Paid by Containment Owner without Receipt	
Cost of Disposal	
Place of Disposal	

Fig. 52: Application Action – Emptying Service Details

Feedback Details

- Click on the Feedback **Detail** button in the Action column of the Application List
- Fill the necessary details and click **Save** button
- Click **Back to List** button if the user wants to go back to Application List without saving

Feedback Details

Back to List

Application ID

Emptied Date

Feeling at the time of emptying

Comments

Back to List **Save**

Fig. 53: Application Action – Feedback Details

7.3.3 Tools

There is a single tool **Export to Excel**. It is used for exporting records in Application List page in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Application List.

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

id	application_date	payment	containcd	approved	assessment	emptying	feedback	holding	customer	father_mc	husband_v	ward	road
25			C00270										7
42	10/24/2019		C02928					XXXX-XX	Shailesh St	shailesh sh	shailesh sh		1 Kathmar
33	11/21/2018		C00047			1	1	1 123456	Araman	Amir			5
11			C00045			1	1	1					7
12	11/18/2018		C00010			1	1	1	Ishlam Mehaboob				7
13			C00008			1	1						7

Fig. 54: Application - Export to Excel tool

7.4 Emptying Service History

This module maintains a list of containments that have been emptied. There are two different filters available that enables user to filter by Containment Code and Year. E.g. the user can filter the list for a particular year,

- Navigate to FSM Info Management > Emptying Service History

Emptying Service History

Show entries

Containment Code	Emptying Date	Start Time	End Time	Uploaded By	Actions
C00042				bhusan	
C00045	2019-07-15			bhushan	
C00047	2018-11-24	12:24:12	02:24:16	sichu	
C00091	2019-09-18	12:19:00	12:19:00	bhusan	
C00283	2019-10-09			karan	
C10000	2018-11-18	03:36:46	04:36:49	sichu	
C10000	2018-11-18	02:38:06	05:38:09	sichu	

Fig. 55: Emptying Service History

7.4.1 Filters

- Choose an appropriate Filter as desired, select its appropriate value and click on **Filter** button. To remove filter, click **Reset** button.

Show entries

Containment Code	Emptying Date	Start Time	End Time	Uploaded By	Actions
C00045	2019-07-15			bhushan	
C00091	2019-09-18	12:19:00	12:19:00	bhusan	
C00283	2019-10-09			karan	

Fig. 56: Using filters to find list of containment emptied in a particular year

7.4.2 Actions

There is a single Actions available for each record. Use **Details** Action when the user wants to just view attributes.

Table 5: Actions for Emptying Service History

ICON	Name	Task that will be performed
	Detail	Shows detail information about the Emptying service

Detail

- Click on the **Detail** button in the Action column Emptying Service History page

- Explore attributes of Emptying service record as desired
- Click on **Back to List** button to go to Emptying Date List

Tools

Export to Excel is used to export records in Emptying Date List page in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Emptying Date List.

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Next Emptying Date	Containment	Actions
2019-11-05	C02310	
2019-11-05	C02329	
2019-11-05	C02362	
2019-11-05	C02843	
2019-11-05	C03399	
2019-11-05	C03483	

next_emptying_date	contained	ward no	road name	BIN	Contact No	Vacutug type	customer name
11/5/2019	C01463	7	Mohisha Kundu Road	B19755			
11/5/2019	C02310	3		B07469			
11/5/2019	C02329	3		B07194			
11/5/2019	C02362	4	Alhera Para Road	B06850			
11/5/2019	C02843	3		B07767			
11/5/2019	C03399	1		B04851			
11/5/2019	C03483	1	Shahid Mamun Road (Gobin*	B02523			
11/5/2019	C03957	1		B00914			
11/5/2019	C05610	8	Chan Para Road	B22097			
11/5/2019	C07534	5	Abul Hossain Road	B12242			

Fig. 57: Emptying Dates - Export to Excel tool

7.5 Next Emptying Info

This module is for efficient planning and optimum use of available resources for optimal functioning of FSM. The user can clearly see with a graph, that how many containment emptying requests could come by weekly basis. There are different filters available that enables user to filter by different time frame. E.g. the user can filter the list for a particular month in a year,

- Navigate to FSM Info Management > Next Emptying Info

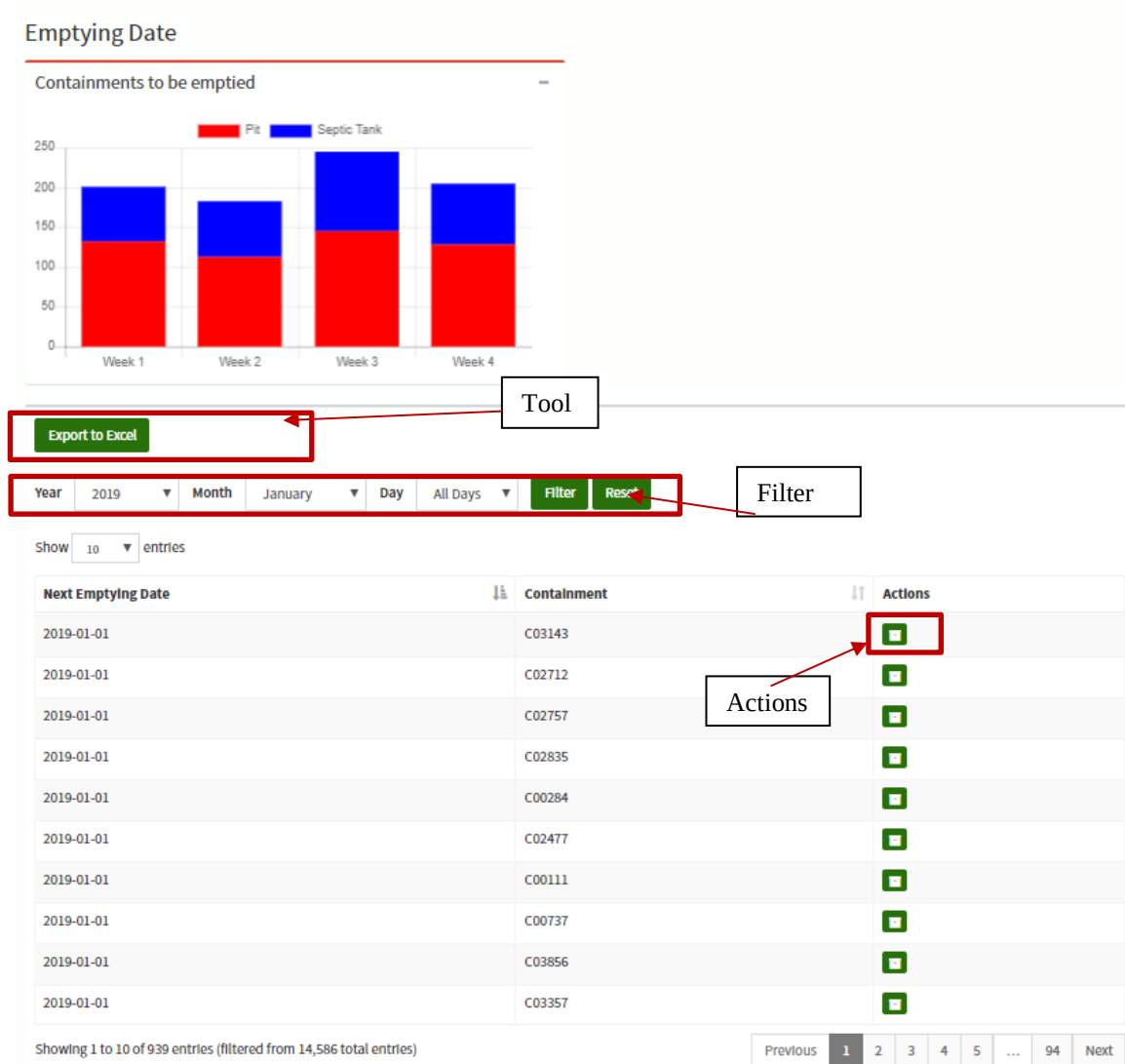


Fig. 58: Potential Emptying Dates

7.5.1 Filters

- Choose an appropriate Filter as desired, select its appropriate value and click on Filter button. To remove filter, click Reset.

Export to Excel

Year 2019 ▼ Month November ▼ Day 5 ▼ Filter Reset

Show 10 ▼ entries

Next Emptying Date	Containment	Actions
2019-11-05	C02310	
2019-11-05	C02329	
2019-11-05	C02362	
2019-11-05	C02843	
2019-11-05	C03399	
2019-11-05	C03483	
2019-11-05	C03957	

Fig. 59: Using filters to find probable containment to be emptied at particular date

7.5.2 Actions

There is a single Actions available for each record. Use **Details** Action when the user wants to just view attributes.

Table 6: Actions for Next Emptying Info

ICON	Name	Task that will be performed
	Detail	Shows detail information about the Next Emptying Date records

Detail

- Click on the **Detail** button in the Action column Emptying Date List
- Explore attributes of records as desired
- Click on **Back to List** button to go to Emptying Date List

Tools

Export to Excel is used to export records in Emptying Date List page in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Emptying Date List.

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer

- View the downloaded file in Excel

Export to Excel

Year: 2019 Month: November Day: 5 Filter Reset

Show 10 entries

Next Emptying Date	Containment	Actions
2019-11-05	C02310	[Icon]
2019-11-05	C02329	[Icon]
2019-11-05	C02362	[Icon]
2019-11-05	C02843	[Icon]
2019-11-05	C03399	[Icon]

next_emptying_date	containcd	ward no	road name	BIN	Contact No	Vacutug type	customer name
11/5/2019	C01463	7	Mohisha Kundu Road	B19755			
11/5/2019	C02310	3		B07469			
11/5/2019	C02329	3		B07194			
11/5/2019	C02362	4	Alhera Para Road	B06850			
11/5/2019	C02843	3		B07767			
11/5/2019	C03399	1		B04851			
11/5/2019	C03483	1	Shahid Mamun Road (Gobin*	B02523			
11/5/2019	C03957	1		B00914			
11/5/2019	C05610	8	Chan Para Road	B22097			
11/5/2019	C07534	5	Abul Hossain Road	B12242			

Fig. 60: Emptying Dates - Export to Excel tool

7.6 Transfer Stations

Maintains database of Transfer Stations. Transfer Stations are collections point for buildings that are not accessible by vehicle. Containment sludge of buildings that are not accessible by vehicle are first transferred to Transfer Station and later Vacutug will be used for emptying the transfer station.

- Navigate to FSM Info Management > Transfer Stations

Transfer Station List

Create new Transfer Station Export to Excel

Show 10 entries

Search:

Transfer Station Code	Name	Longitude	Latitude	Actions
TS00001	Transfer Station 1			[Icon]
TS00002	Transfer Station 2			[Icon]
TS00003	Transfer Station 3			[Icon]
TS00004	Transfer Station 4			[Icon]
TS00005	Transfer Station 5			[Icon]

Fig. 61: Transfer Station List

7.6.1 Filters

- Type the search term in the search filter. The list will be dynamically updated to match search term
- Delete search term to get back to the previous list

Show entries

Search

Transfer Station Code	Name	Longitude	Latitude	Actions
TS00001	Transfer Station 1			   

Fig. 62: Using search filter to find a station with particular Transfer Station Code

7.6.2 Actions

There are four different Actions available for each record. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there a need of modifying any particular attribute.

Table 7: Actions for User List

ICON	Name	Task that will be performed
	Edit	Opens form to edit the Treatment Plant record details.
	Detail	Shows detail information about the Treatment Plant record
	Delete	Deletes a Treatment Plant record
	Map	Shows spatial location of the Treatment Plants on a map

Edit

- Click on the **Edit** button in the Action column of the Transfer Station List
- In Edit Transfer Station page, enter empty fields or change/modify already added fields as necessary
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Transfer Station List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Service Transfer Station List page

Edit Transfer Station

Transfer Station Code: TS00001

Name: Transfer Station 1

Description:

Back to List Save

Fig. 63: Transfer Station Action - Edit

Detail

- Click on the **Detail** button in the Action column of the Transfer Station List
- Explore attributes of Treatment Plant record as desired
- Click on **Back to List** button to go to Transfer Station List

Delete

- Click on the **Delete** button in the Action column of the Treatment Plant List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

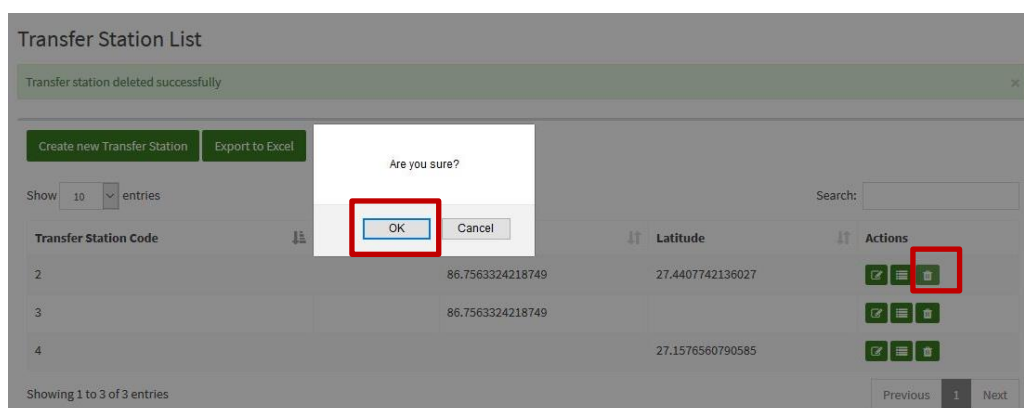


Fig. 64: Transfer Stations Action - Delete

Map

- Click on **Map** button in the Action column of the Transfer Station List. A map will open, showing a spatial location of the Transfer Station with a marker
- To go back to Transfer Station page, either click Back button on the browser or navigate to FSM Info Management > Transfer Stations.

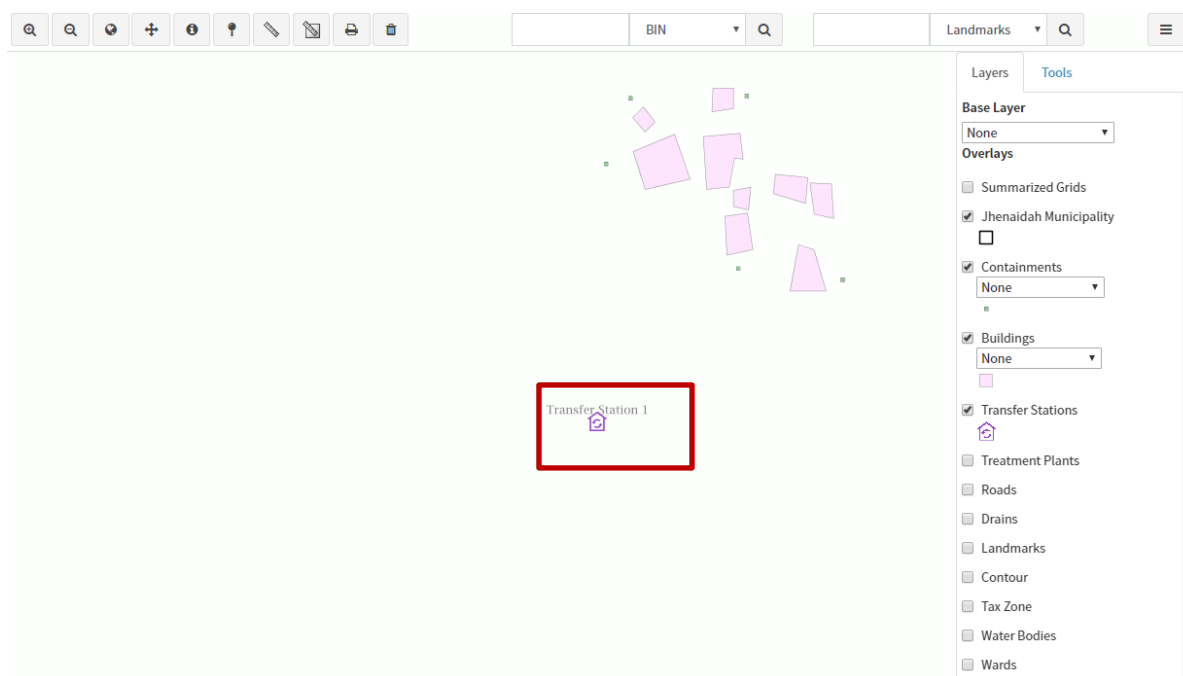


Fig. 65: Treatment Plant Action: Map

7.6.3 Tools

There are two tools in the Transfer Station List page. **Create New Transfer Station** and **Export to Excel** tool. **Create New Transfer Station** tool is used to add new Transfer Station record in the Transfer Station List. **Export to Excel** is used to export records in Treatment Plant List in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Transfer Station List.

Create New Transfer Station

- Click on **Create New Transfer Station** button.
- Then fill the required data. Set a location of the Transfer Station by dragging a red marker on the map. Notice that Longitude and Latitude fields are automatically filled as the user drag red marker on the bottom map. Click **Save** button.
- Newly added Transfer Plant record will be added at the end of the list

Create Transfer Station

Transfer Station Code

Name

Longitude

Latitude

Description

Drag Marker To Set Latitude And Longitude



Fig. 66: Transfers Stations - Create New Transfer Station tool

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Create new Transfer Station

Show entries

Transfer Station Code	Name	Longitude	Latitude	Actions
TS00001	Transfer Station 1			Edit Delete Location

	A	B	C	D	E	F
1	transstnc	name	longitude	latitude	description	
2	TS00001	Transfer Station 1				
3						

Fig. 67: Transfer Stations - Export to Excel tool

7.7 Treatment plants

Maintains database of treatment plants that will receive sludge. Treatment plants performs treatment on sludge and converts to Compost.

- Navigate to FSM Info Management > Treatment Plants

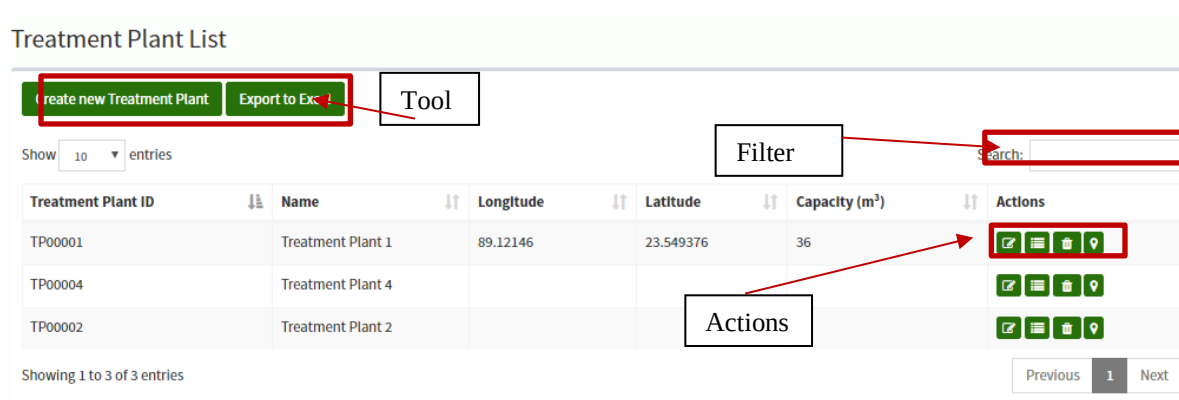


Fig. 68: Treatment Plant List

7.7.1 Filters

- Type the search term in the search filter. The list will be dynamically updated to match search term
- Delete search term to get back to the previous list

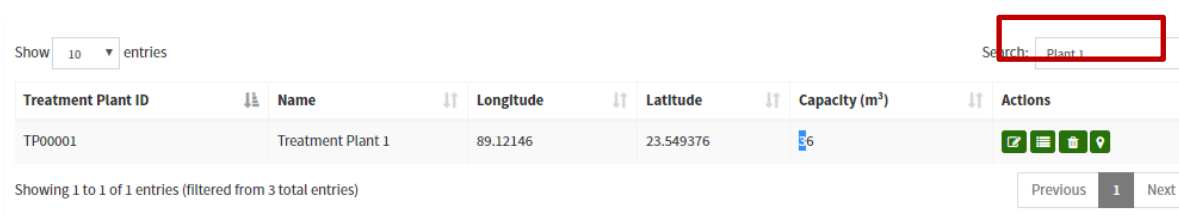


Fig. 69: Using search filter to find Treatment Plant 1

7.7.2 Actions

There are four different Actions available for each record. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there a need of modifying any particular attribute.

Table 8: Actions for User List

ICON	Name	Task that will be performed

	Edit	Opens form to edit the Treatment Plant record details.
	Detail	Shows detail information about the Treatment Plant record
	Delete	Deletes a Treatment Plant record
	Map	Shows spatial location of the Treatment Plants on a map

Edit

- Click on the **Edit** button in the Action column of the Treatment Plant List
- In Edit Treatment Plant page, enter empty fields or change/modify already added fields as necessary.
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Treatment Plant List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Service Treatment Plant List page

Edit Treatment Station

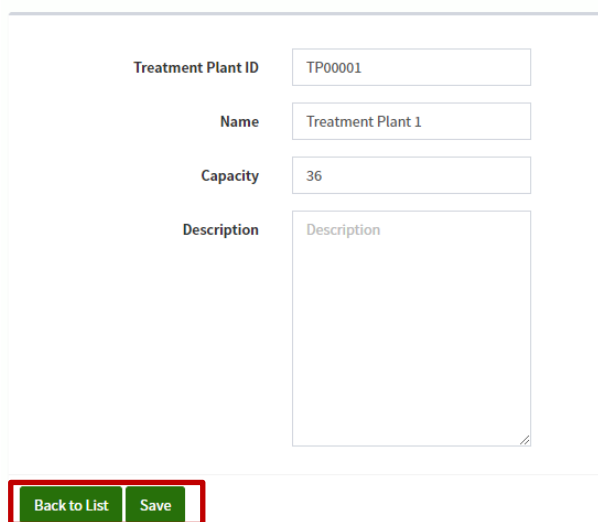


Fig. 70: Treatment Plants Action - Edit

Detail

- Click on the **Detail** button in the Action column of the Treatment Plant List
- Explore attributes of Treatment Plant record as desired
- Click on **Back to List** button to go to Treatment Plant List

Delete

- Click on the **Delete** button in the Action column of the Treatment Plant List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

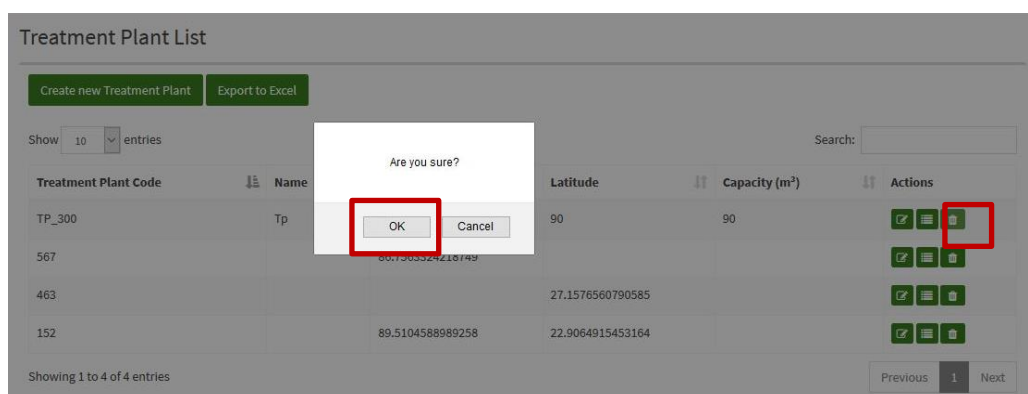


Fig. 71: Treatment Plants Action - Delete

Map

- Click on **Map** button in the Action column of the Treatment Plant list. A map will open, showing a spatial location of the Treatment Plant with a marker
- To go back to Treatment Plant List page, either click Back button on the browser or navigate to FSM Info Management > Treatment Plants



Fig. 72: Treatment Plant Action: Map

7.7.3 Tools

There are two tools in the Treatment Plant List page. **Create New Treatment Plant** and **Export to Excel** tool. **Create New Treatment Plant** tool is used to add new Treatment Plant record in the Treatment Plant List. **Export to Excel** is used to export records in Treatment Plant List in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Treatment Plant List.

Create New Treatment Plant

- Click on **Create New Treatment Plant** button.
- Then fill the required data. Set a location of the Treatment Plant by dragging a red marker on the map. Notice that Longitude and Latitude fields are automatically filled as the user drag red marker on the bottom map. Click **Save** button.
- Newly added Treatment Plant record will be added at the end of the list

Create Treatment Plant

Treatment Plant ID

Name

Longitude

Latitude

Capacity

Description

Drag Marker To Set Latitude And Longitude



Fig. 73: Treatment Plants - Create New Treatment Plant tool

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Create new Treatment Plant

Show entries

Search:

Treatment Plant ID	Name	Longitude	Latitude	Capacity (m³)	Actions
TP00001	Treatment Plant 1	89.12146	23.549376	36	Edit Delete Add

	A	B	C	D	E	F	G
1	trtpltcd	name	longitude	latitude	capacity	description	
2	TP00001	Treatment	89.12146	23.549376	36		
3							

Fig. 74: Treatment Plant Providers - Export to Excel tool

7.8 Service providers

Maintains database of service providers that provide containment emptying service.

- Navigate to FSM Info Management > Service Providers

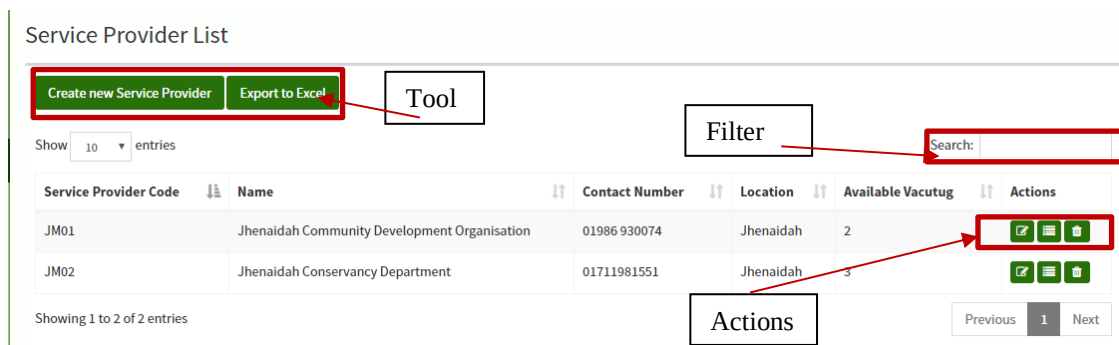


Fig. 75: Service Providers List

7.8.1 Filters

- Type the search term in the search filter. The list will be dynamically updated to match search term
- Delete search term to get back to the previous list

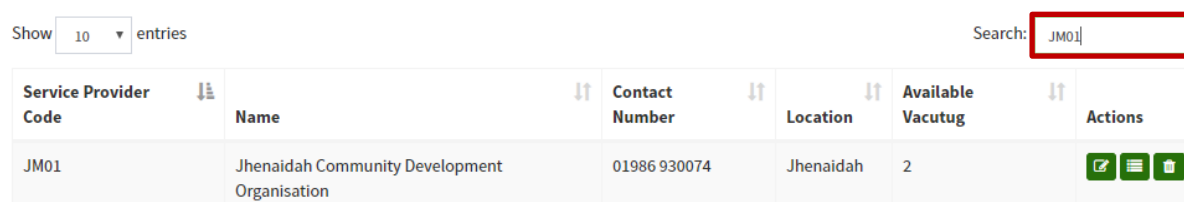


Fig. 76: Using search filter to find Service Provider JM01

7.8.2 Actions

There are three different Actions available for each record. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there a need of modifying any particular attribute.

Table 9: Actions for User List

ICON	Name	Task that will be performed
	Edit	Opens form to edit the Service Provider record details.
	Detail	Shows detail information about the Service Provider record
	Delete	Deletes a Service Provider record

Edit

- Click on the **Edit** button in the Action column of the Service Provider List
- In Edit Service Provider page, enter empty fields or change/modify already added fields as necessary.
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Service Providers List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Service Provider List page

Edit Service Provider

Service Provider Code	JM01
Name	Jhenaidah Community Development Organisation
Contact Name	Munna
Contact Number	01986 930074
Location	Jhenaidah
Ward	1 ▼
Available Vacutug	2
Email	jm01@gmail.com

[Back to List](#)
[Save](#)

Fig. 77: Service Providers Action - Edit

Detail

- Click on the **Detail** button in the Action column of the Service Provider List
- Explore attributes of Service Provider record as desired
- Click on **Back to List** button to go to Service Provider List

Delete

- Click on the **Delete** button in the Action column of the Service Provider List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

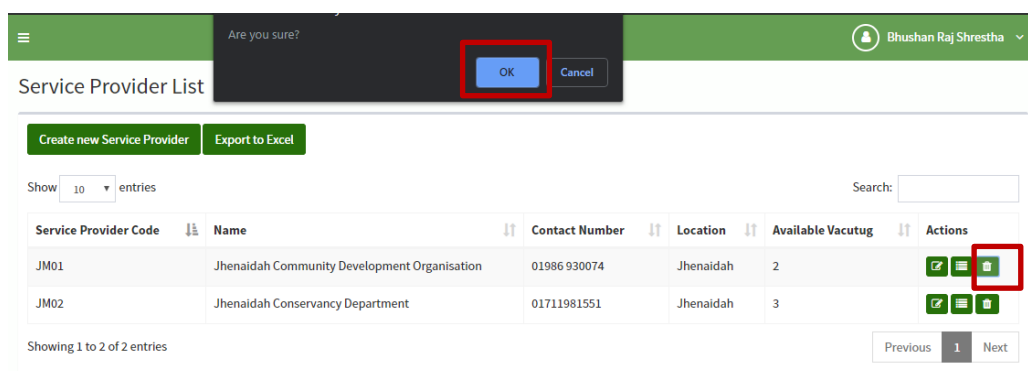


Fig. 78: Servicer Provider Action - Delete

7.8.3 Tools

There are two tools in the Service Providers List page. **Create New Service Provider** and **Export to Excel** tool. **Create New Service Provider** tool is used to add new Service Provider record in the Service Provider List. **Export to Excel** is used to export records in Service Provider Sale List in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Service Provider List.

Create New Service Provider

- Click on **Create New Service Provider Sale** button.
- Then fill the required data and Click **Save** button
- Newly added Service Provider record will be added at the end of the list

Create Service Provider

The screenshot shows the 'Create Service Provider' form. It has several input fields: 'Service Provider Code', 'Name', 'Contact Name', 'Contact Number', 'Location', 'Ward' (a dropdown menu), 'Available Vacutug', and 'Email'. At the bottom of the form, there are two buttons: 'Back to List' and 'Save' (highlighted with a red box).

Fig. 79: Service Providers - Create New Service Provider tool

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Create new Service Provider Export to Excel

Show 10 entries Search: JM01

Service Provider Code	Name	Contact Number	Location	Available Vacutug	Actions
JM01	Jhenaidah Community Development Organisation	01986 930074	Jhenaidah	2	  

1	A	B	C	D	E	F	G	H	I	J
	servcode	name	contnm	offtel	location	ward	vactugavl	email		
2	JM01	Jhenaidah	Munna	01986 930	Jhenaidah	1 2		jm01@gmail.com		
3										
4										

Fig. 80: Service Providers - Export to Excel tool

7.9 Compost Sales

Maintains database of sale of compost by treatment plants. A daily compost sale log book is maintained by treatment plants. Subsequently compost sale database is updated.

- Navigate to FSM Info Management > Compost Sales

Compost Sale List

Create new Compost Sale Export to Excel

Show 10 entries

Filter Search:

Date	Treatment Plant	Weight	Amount	Actions
2018-11-01	Treatment Plant 1	10	1000	  
2019-06-14	Treatment Plant 2	100	1000	  
2019-06-20	Treatment Plant 4	24	500	  

Showing 1 to 3 of 3 entries

Previous 1 Next

Fig. 81: Compost Sale List

7.9.1 Filters

- Type the search term in the search filter. The list will be dynamically updated to match search term
- Delete search term to get back to the previous list

Show 10 entries Search: Plant 1

Date	Treatment Plant	Weight	Amount	Actions
2018-11-01	Treatment Plant 1	10	1000	  

Showing 1 to 1 of 1 entries (filtered from 3 total entries)

Previous 1 Next

Fig. 82: Using search filter to find Compost Sales by Treatment Plant1

7.9.2 Actions

There are three different Actions available for each record. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there a need of modifying any particular attribute.

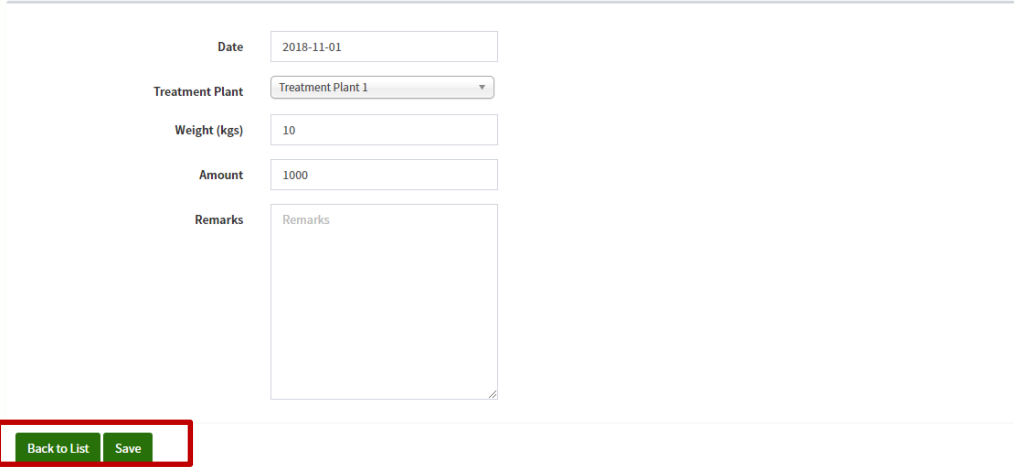
Table 10: Actions for Compost Sale

ICON	Name	Task that will be performed
	Edit	Opens form to edit the Compost Sales record details.
	Detail	Shows detail information about the Compost Sales record
	Delete	Deletes a Compost Sales record

Edit

- Click on the **Edit** button in the Action column of the Compost Sales List
- In Edit Compost Sale page, enter empty fields or change/modify already added fields as necessary.
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Compost Sales List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Compost Sales List page

Edit Compost Sale



The screenshot shows the 'Edit Compost Sale' form with the following fields and values:

- Date: 2018-11-01
- Treatment Plant: Treatment Plant 1
- Weight (kgs): 10
- Amount: 1000
- Remarks: (empty text area)

At the bottom of the form, there are two buttons: 'Back to List' and 'Save'. These buttons are highlighted with a red rectangle.

Fig. 83: Compost Sales Action - Edit

Detail

- Click on the **Detail** button in the Action column of the Compost Sales List
- Explore attributes of Compost Sales record as desired
- Click on **Back to List** button to go to Compost Sales List

Delete

- Click on the **Delete** button in the Action column of the Compost Sales List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

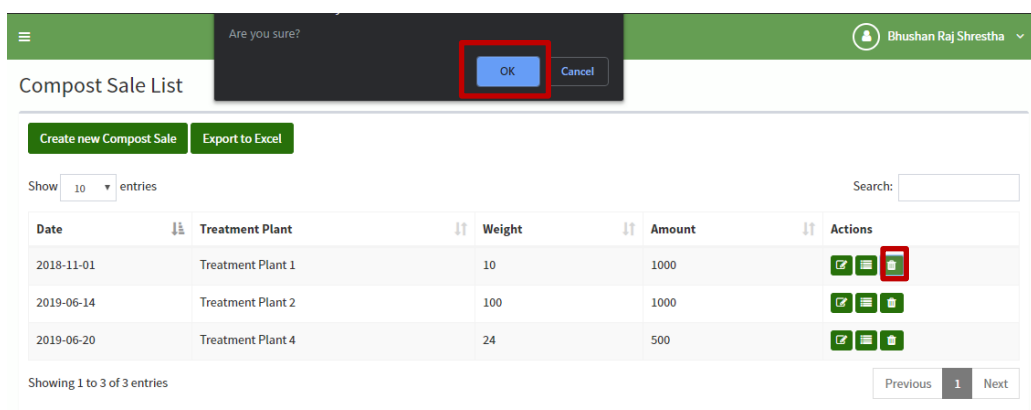


Fig. 84: Compost Sales Action - Delete

7.9.3 Tools

There are two tools in the Compost Sales List page. **Create New Compost Sale** and **Export to Excel** tool. **Create New Compost Sale** tool is used to add new compost sale record in the Compost Sale List. **Export to Excel** is used to export records in Compost Sale List in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Compost Sale List.

Create New Compost Sale

- Click on **Create New Compost Sale** button.
- Then fill the required data and Click **Save** button
- Newly added compost sale record will be added at the end of the list

Create Compost Sale

Date

Treatment Plant

Weight (kgs)

Amount

Remarks

[Back to List](#) [Save](#)

Fig. 85: Compost Sales - Create New Compost Sale tool

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Create new Compost Sale [Export to Excel](#)

Show entries

Search:

Date	Treatment Plant	Weight	Amount	Actions
2018-11-01	Treatment Plant 1	10	1000	Edit Delete

Showing 1 to 1 of 1 entries (filtered from 3 total entries)

Previous **1** Next

	A	B	C	D	E	F
1	date	treatmen	weight	amount	remarks	
2	2018-11-01	Treatment	10	1000		
3						

Fig. 86: Compost Sales - Export to Excel tool

7.10 Sludge Collection

Maintains database of sludge disposal at treatment plants. A daily sludge collection log book is maintained by treatment plants. Subsequently sludge collection database is updated.

- Navigate to FSM Info Management > Sludge Collection

Sludge Collection List

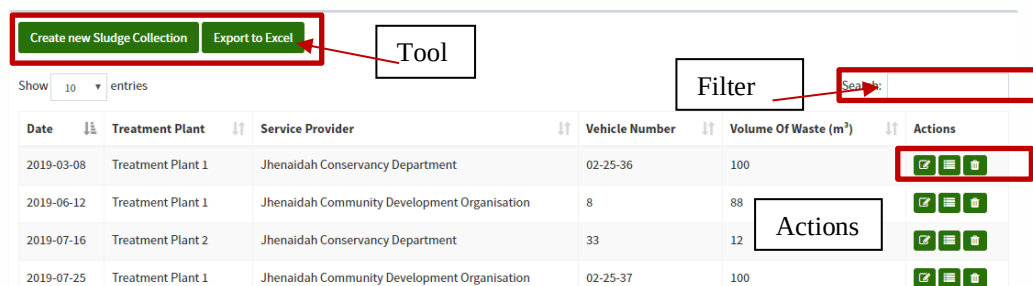


Fig. 87: Sludge Collection List

7.10.1 Filters

- Type the search term in the search filter. The list will be dynamically updated to match search term
- Delete search term to get back to previous list

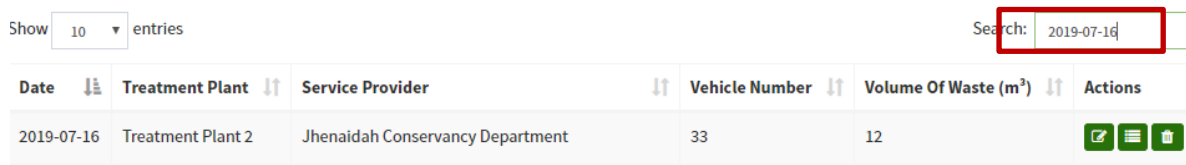


Fig. 88: Using search filter to find sludge collected at particular date

7.10.2 Actions

There are three different Actions available for each record. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there a need of modifying any particular attribute.

Table 11: Actions for Sludge Collection

ICON	Name	Task that will be performed
	Edit	Opens form to edit the Sludge Collection record details.
	Detail	Shows detail information about the Sludge Collection record
	Delete	Deletes a Sludge Collection record

Edit

- Click on the **Edit** button in the Action column of the Sludge Collection List

- In Edit Sludge Collection page, enter empty fields or change/modify already added fields as necessary. Notice that more than one Application ID can be entered in Application ID field.
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Sludge Collection List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Sludge Collection List page

Edit Sludge Collection

Fig. 89: Sludge Collection Action - Edit

Detail

- Click on the **Detail** button in the Action column of the Sludge Collection List
- Explore attributes of Sludge collection record as desired
- Click on **Back to List** button to go to Sludge Collection List

Delete

- Click on the **Delete** button in the Action column of the Sludge Collection List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

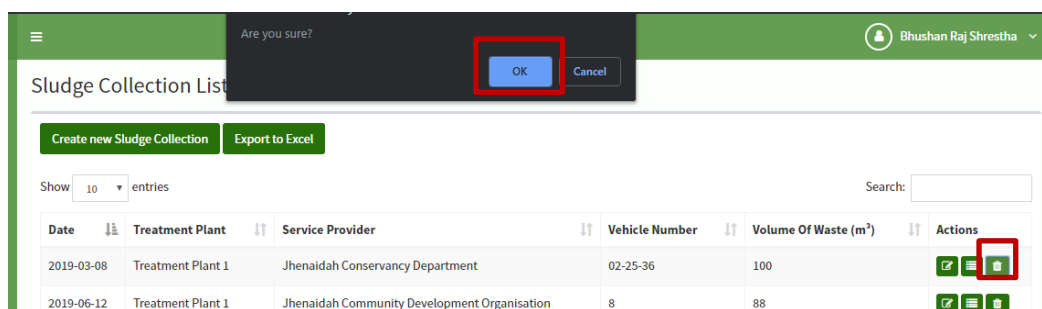


Fig. 90: Sludge Collection Action - Delete

7.10.3 Tools

There are two tools in the Sludge Collection List page. **Create New Sludge Collection** and **Export to Excel** tool. **Create New Sludge Collection** tool is used to add new sludge collection record in the Sludge Collection List. **Export to Excel** is used to export records in Sludge Collection List page in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Sludge Collection List.

Create New Sludge Collection

- Click on **Create New Sludge Collection** button.
- Then fill the required data and Click **Save** button
- Newly added sludge collection record will be added at the end of list

Create Sludge Collection

Date	Date (yyyy-mm-dd)
Time	Time
Treatment Plant	-- Choose treatment plant --
Service Provider	-- Choose service provider --
Vehicle Number	Vehicle Number
Volume Of Waste (m³)	Volume Of Waste (m³)
Application ID	Select Some Options

Fig. 91: Sludge Collection - Create New Sludge Collection tool

Export to Excel

- Click **Export to Excel** button. An Excel file will be download in user's computer
- View the downloaded file in Excel

Create new Sludge Collection

Export to Excel

Show 10 entries

Search: 2019-07-16

Date	Treatment Plant	Service Provider	Vehicle Number	Volume Of Waste (m³)	Actions
2019-07-16	Treatment Plant 2	Jhenaidah Conservancy Department	33	12	

A	B	C	D	E	F	G
1	date	time	treatmen	service_p	vehicle_n	vol_of_waste
2	2019-07-16	18:08:00	Treatment	Jhenaidah	33	12
3						

Fig. 92: Sludge Collection- Export to Excel tool

7.11 Help Desks

Maintains database of Help Desks responsible for overall management of FSM.

- Navigate to FSM Info Management > Help Desks

Help Desks List

Create new Help Desk

Tools

Tools

Show 10 entries

Search:

Help Desk ID	Name	Description	Actions
1	Help Desk 1	desk 1 description.....	
2	Help Desk 2	Help desk description.....	

Showing 1 to 2 of 2 entries

Previous

1

Next

Fig. 93: Help Desks List

7.11.1 Filters

- Type the search term in the search filter. The list will be dynamically updated to match search term
- Delete search term to get back to the previous list

Show	10	entries	Search: Desk 1		
Help Desk ID	Name	Description	Actions		
1	Help Desk 1	desk 1 description.....	 		

Fig. 94: Using search filter to find Help Desk 1

7.11.2 Actions

There are two different Actions available for each record.

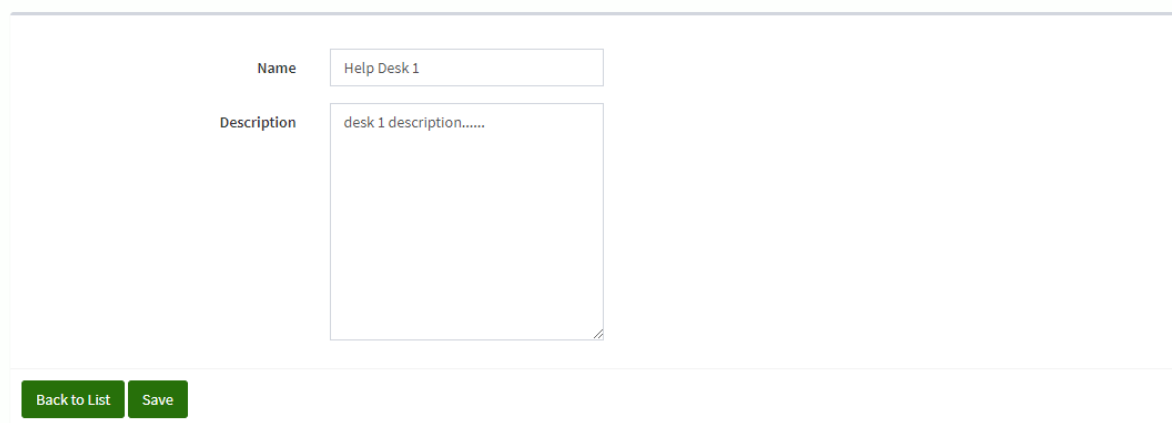
Table 12: Actions for Help Desk List

ICON	Name	Task that will be performed
	Edit	Opens form to edit the Help Desk record details.
	Delete	Deletes a Help Desk record

Edit

- Click on the **Edit** button in the Action column of the Help Desk List
- In Edit Help Desk page, enter empty fields or change/modify already added fields as necessary.
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Help Desk List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Help Desk List page

Edit Help Desk

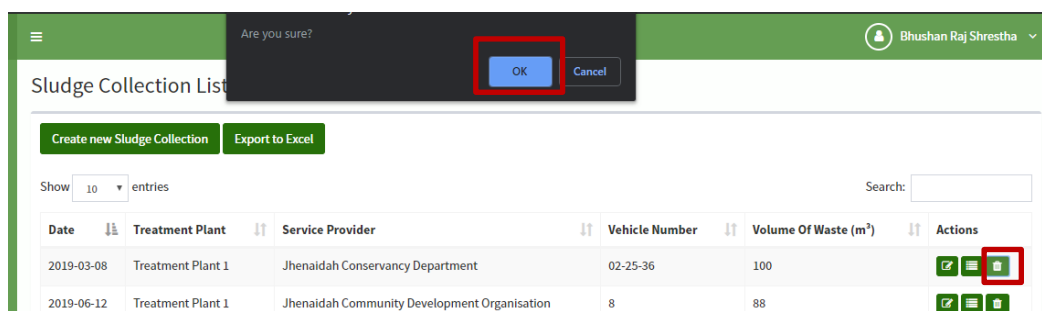


The screenshot shows a web form titled 'Edit Help Desk'. It contains two labeled input fields: 'Name' and 'Description'. The 'Name' field contains the text 'Help Desk 1'. The 'Description' field contains the text 'desk 1 description.....'. Below these fields, there are two green buttons: 'Back to List' and 'Save'.

Fig. 95: Help Desk Action - Edit

Delete

- Click on the **Delete** button in the Action column of the Help Desk List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.



Help Desk Action - Delete

7.11.3 Tools

Create New Sludge Collection tool is used to add new sludge collection record in the Sludge Collection List. **Export to Excel** is used to export records in Sludge Collection List page in an Excel file. It is to be noted that **Export to Excel** works seamlessly with search filter available in the Sludge Collection List.

Create New Help Desk

- Click on **Create New Help Desk** button.
- Then fill the required data and Click **Save** button
- Newly added Help Desk record will be added at the end of list

Create Help Desk

Name

Description

Back to List

Save

Fig. 96: Create New Help Desk tool

8 Data Export

This is a module dedicated to support further planning and analysis outside the IMIS system. All data layers stored in the IMIS system is available for export. There is an option to export layers within a specific ward or tax zones. Layers can be exported in either shape file or KML file. Use any GIS software to view exported shape file. KML file can be easily viewed in Google Earth.

Export To Shape File Or Kml

Layers

☒ Wards ☐ Taxzones

Wards

Export Format

--- Select a Format ---

Download

Fig. 97: Data Export Functionally

- Enter a specific data layer that the user is interested to export. The checklist available in Layers filter is not mutually exclusive, therefore the user has an option of selecting all layers, a specific data layer or select multiple data layers.
- Choose a desired spatial Filter between Wards and Tax zones. Depending upon the chosen spatial Filter, the user is provided with an option will be available to select Wards/ Tax zones. The checklist available in Wards/ Tax zones filter is not mutually exclusive, therefore the user can select more than one Wards/ Tax zones if desired.
- Choose a desired export format between shape file and KML file. A zipped file will be downloaded when shape file is chosen.
- Click **Download** button

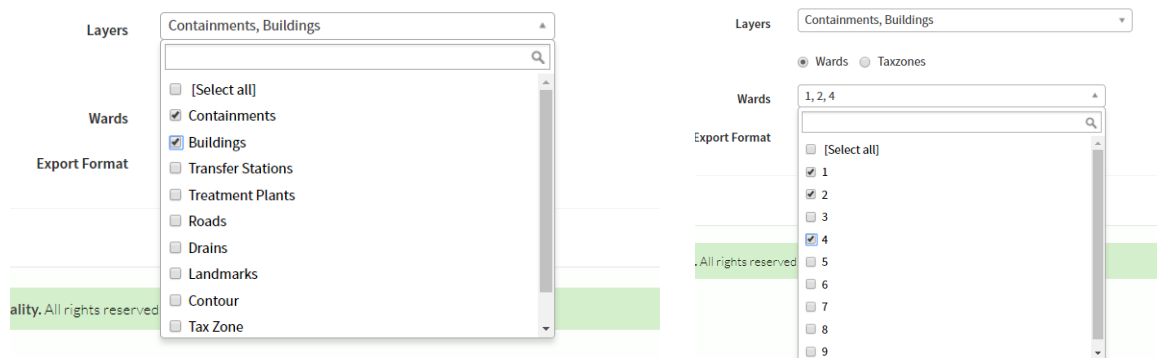


Fig. 98: Examples of not mutually exclusive filters in Data Export

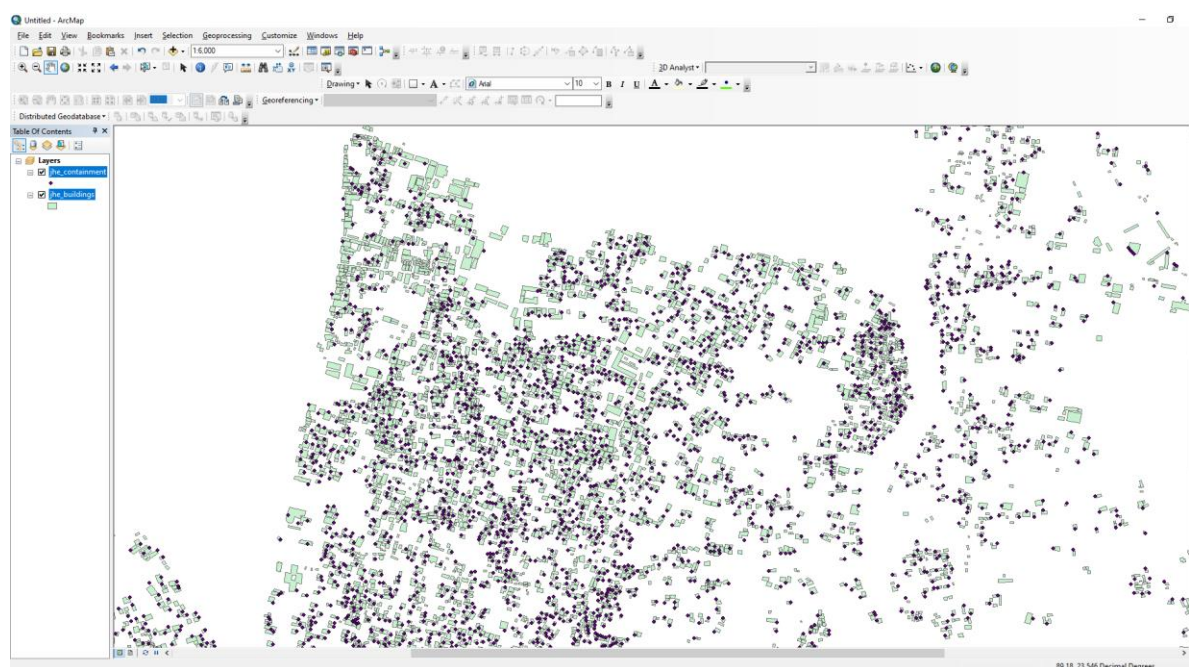


Fig. 99: Viewing exported shape files in ArcGIS

9 Data Import

This is a module dedicated to bulk import/upload users' payment info in "CSV" format. As of now, two types of data import are supported in the system:

1. Revenue (Tax payments)
2. Water Supply Payments

Import CSV Tax Payments

Click here to import from excel

Export to excel

Ward

All Ward

Due

All Due

Match

All

Filter

Reset

Show

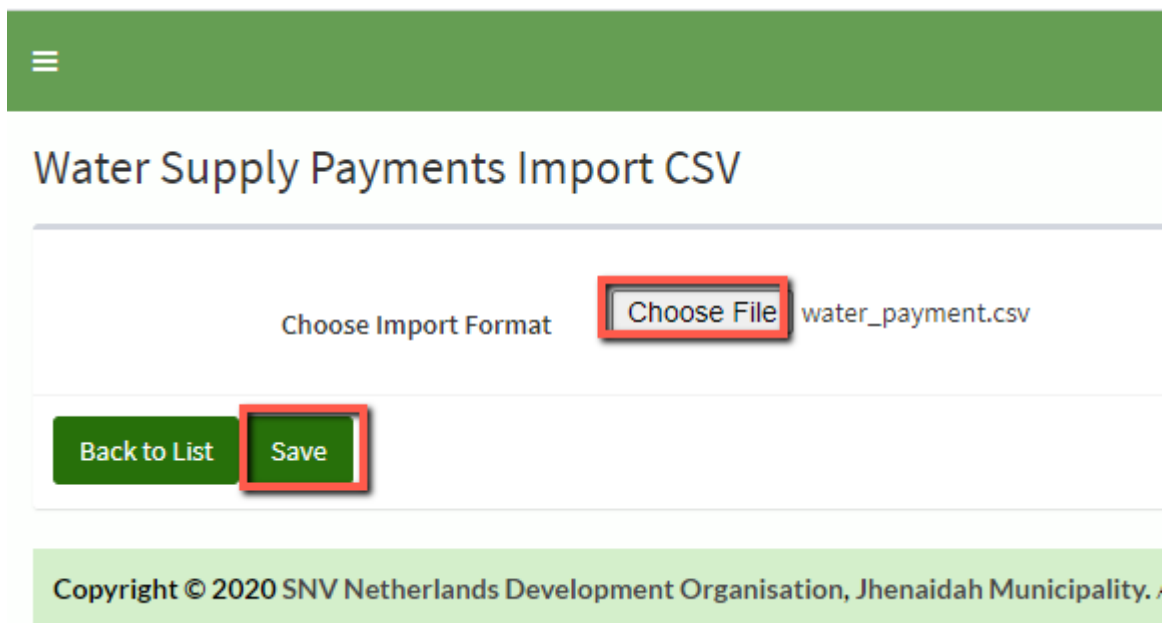
10

entries

Tax Code	Owner	Years Due	Ward	Match
01-999-9999-99		No Due		N
01-004-0175-00	Test Owner	1 Year	1	Y
01-004-0175-00	Test Owner	1 Year	1	Y
01-004-0175-00	Test Owner	1 Year	1	Y

Fig. 100: Table of existing tax information

- Navigate to either Data Import > Revenue or Data Import > Water Supply Payments
- A table of existing tax information for the corresponding service will be displayed.
- User can Filter the table by Ward, Due Date or Match
- Click on "**Click here to import from excel**" button to load the import data interface
- Click on "Choose File" button and select an appropriate CSV file from computer
- Click the "**Save**" button to import the payment data. Valid data will be available in the tax table.



The image shows a web application interface for importing CSV data. At the top, there is a green header bar with a white hamburger menu icon. Below the header, the title "Water Supply Payments Import CSV" is displayed in a large, dark font. The main content area is a light green box containing a form. The form has a label "Choose Import Format" followed by a file selection button labeled "Choose File" and a text input field containing "water_payment.csv". Below the form, there are two green buttons: "Back to List" and "Save". The "Save" button is highlighted with a red border. At the bottom of the interface, there is a green footer bar with the text "Copyright © 2020 SNV Netherlands Development Organisation, Jhenaidah Municipality."

Water Supply Payments Import CSV

Choose Import Format water_payment.csv

Copyright © 2020 SNV Netherlands Development Organisation, Jhenaidah Municipality.

Fig. 101: Data Import Interface

10 Settings

This is a module dedicated to support maintaining user list, defining roles of the users and providing/limiting permissions to the users what they can or can't do in IMIS system.

10.1 Users

- Navigate to Settings > Users

Users Lists

Create new User Tools Filters Search:

Show 10 entries

Email	Username	Name	Role	Actions
ershad@gmail.com	ershad123	Ershad Ali	Help Desk	  
a2i@a.com	a2i_user	A2i User	A2i	  
supervisor@gmail.com	supervisor	Supervisor	Ward User, Supervisor, Emptying Service, Sludge Collection User, Compost Sale User	  
naresh@gmail.com	naresh	Naresh Shrestha	Ward User, Emptying Service, Sludge Collection User, Compost Sale User	  
avein@biztechnepal.com	avein	Avein Tamrakar	Compost Sale User	  
prasanna@biztechnepal.com	prasanna	Prasanna Tuladhar	Sludge Collection User	  
mcasas@snv.org	marcpc	Marc Perez Casas	Admin	  
sundangol@gmail.com	suraj	Suraj Dangol	Admin	  
shrestha.sichu@gmail.com	sichu	Sichu Shrestha	Ward User	  
ankitashakya29@gmail.com	ankita	Ankita Shakya	Ward User	  

Showing 1 to 10 of 14 entries

Previous 1 2 Next

Fig. 102: User List

Attributes shown can be sorted in ascending/descending order. There is a search filter available that enables user to filter the User List according to their need. E.g. look for users assigned with Ward User roles only. There are different Actions buttons available for each record. The user has a flexibility to choose how many records (10, 25, 50 & 100) are displayed on a page.

10.1.1 Filters

- Type the search term in the search filter. The list will be dynamically updated to match search term
- Delete search term to get back to the previous list

Show 10 entries

Search: Ward user

Email	Username	Name	Role	Actions
supervisor@gmail.com	supervisor	Supervisor	Ward User, Supervisor, Emptying Service, Sludge Collection User, Compost Sale User	  
naresh@gmail.com	naresh	Naresh Shrestha	Ward User, Emptying Service, Sludge Collection User, Compost Sale User	  
shrestha.sichu@gmail.com	sichu	Sichu Shrestha	Ward User	  
ankitashakya29@gmail.com	ankita	Ankita Shakya	Ward User	  
sagar@gmail.com	sagar	Sagar Shakya	Ward User, Supervisor	  

Fig. 103: Using search filter to find user assigned to Ward Roles

10.1.2 Actions

There are three different Actions available for each record. Use **Details** Action when the user wants to just view attributes. Use **Edit** Action when there a need of modifying any particular attribute.

Table 13: Actions for User List

ICON	Name	Task that will be performed
	Edit	Opens form to edit the User details.
	Detail	Shows detail information about the User
	Delete	Deletes the user

Action: Edit

- Click on the **Edit** button in the Action column of the User List
- In Edit User page, enter empty fields or change/modify already added fields as necessary
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to User List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to User List page

Edit User

Edit

Full Name	Ershad Ali
Email	ershad@gmail.com
Username	ershad123
Password	Password
Confirm Password	Confirm Password
Role	Help Desk X
Ward	Ward
Treatment Plant	--- Choose treatment plant ---
Help Desk	Help Desk 1

Fig. 104: User Action - Edit User

Action: Details

- Click on the **Detail** button in the Action column of the User list
- Explore attributes of Users as desired
- Click on **Back to List** button to go to User List

User Detail

Full Name	A2i User
Email	A2i@a.com
Username	A2i User
Role	A2i
Ward	

Fig. 105: User Action - Details

Action: Delete

- Click on the **Delete** button in the Action column of the User List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

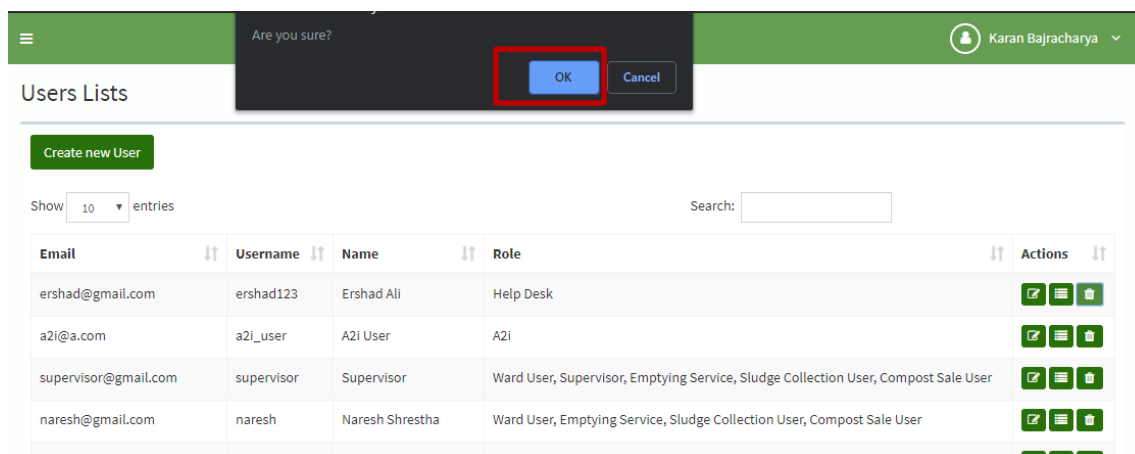


Fig. 106: User Action - Delete

10.1.3 Tools

Add User tool is used to add new User and defined their roles.

Add User

- Click on **Create New User** button.
- Then fill the required data. Define what the user Roles are and Click **Save** button
- Newly added User will be added at the end of list in User List

Create User

Full Name	Full Name
Email	Email
Username	Username
Password	Password
Confirm Password	Confirm Password
Role	Select Some Options
Ward	Ward
Treatment Plant	--- Choose treatment plant ---
Help Desk	--- Choose help desk ---

Fig. 107: Create New User tool

10.2 Roles

- Navigate to Settings > Roles

Attributes shown can be sorted in ascending/descending order. There is a search filter available that enables user to filter the Roles list according to their need. There are different Actions buttons available for each record. The user has a flexibility to choose how many records (10, 25, 50 & 100) are displayed on a page.

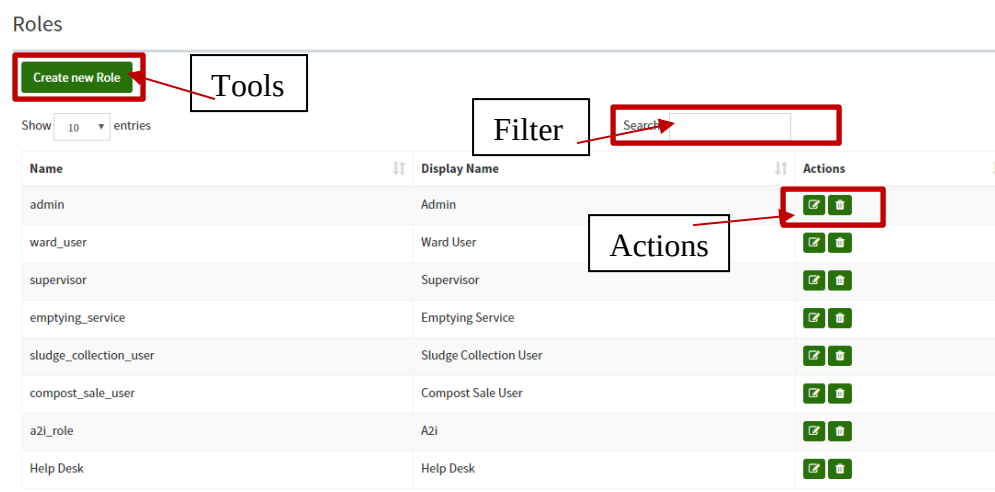


Fig. 108: Roles List

10.2.1 Actions

There are two different Actions available for each record.

Table 14: Actions for Role List

ICON	Name	Task that will be performed
	Edit	Opens form to edit the Role details.
	Delete	Deletes the Role

Action: Edit

- Click on the **Edit** button in the Action column of the Role List
- In Edit Role page, enter empty fields or change/modify already added fields as necessary
- Click on **Save** button if the user wants to save any changes. After saving, the user will be directed to Role List page
- Click on **Back to List** if the user does not want to save any changes. It will direct user to Role List page

Edit Role

Fig. 109: Roles Action: Edit

Action: Delete

- Click on the **Delete** button in the Action column of the Roles List.
- It will display a confirmation dialog. Click OK button in the confirmation dialog.

Name	Display Name	Actions
admin	Admin	[Edit] [Delete]
ward_user	Ward User	[Edit] [Delete]
supervisor	Supervisor	[Edit] [Delete]
emptying_service	Emptying Service	[Edit] [Delete]
sludge_collection_user	Sludge Collection User	[Edit] [Delete]

Fig. 110: Roles Action: Edit

10.2.2 Tools

Create New Role tool is used to add new Role.

Create New Role

- Click on **Create New Role** button.
- Then fill the required data and Click **Save** button
- Newly added Role will be added at the end of list in Role List

Add Role

Fig. 111: Create New Role tool

10.3 Permissions

- Navigate to Settings > Permissions

A Role can be assigned to numbers of functionalities in IMIS. Permission & Restriction to a particular functionality is controlled by Permission page. If any functionality's permission is granted to a Role, all the users that are assigned to that Role will automatically inherit permission to that functionality. There are over 148 functionalities that can be applied to a particular Role.

- To give **Permissions** to the Role **Tick** the Check boxes of the Permissions under the Role
- Click the **Save** button

Permissions

Permission	Admin	Ward User	Supervisor	Emptying Service	Sludge Collection User	Compost Sale User	A2i	Help Desk
List Users	☒	☐	☐	☐	☐	☐	☒	☐
View User	☒	☐	☐	☐	☐	☐	☒	☐
Add User	☒	☐	☐	☐	☐	☐	☐	☐
Edit User	☒	☐	☐	☐	☐	☐	☐	☐
Delete User	☒	☐	☐	☐	☐	☐	☐	☐
List Roles	☒	☐	☐	☐	☐	☐	☒	☐
View Role	☒	☐	☐	☐	☐	☐	☒	☐
Add Role	☒	☐	☐	☐	☐	☐	☐	☐
Edit Role	☒	☐	☐	☐	☐	☐	☐	☐
Delete Role	☒	☐	☐	☐	☐	☐	☐	☐
Assign Permissions	☒	☐	☐	☐	☐	☐	☐	☐

Fig. 112: Granting permissions to Roles

11 View Map

This is the main graphical web interface of IMIS where spatial visualization of data layers and other spatially assisted decision making process will be performed. There are three main sections in the map.

1. General tools and search functionality
2. Map
3. Layers TAB
4. Tools TAB

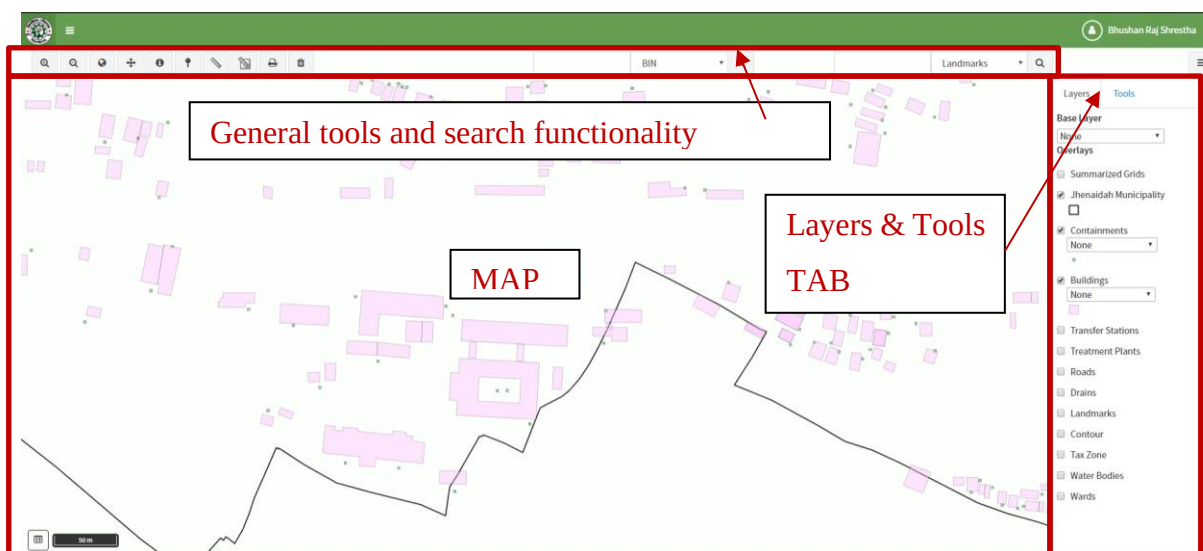


Fig. 113: Map of IMIS

11.1 General tools

General tools and search functionality are located on the header (located in the upper part of the page). These tools are mainly for navigation around the map and performing basic operations e.g. measurement or printing.

<u>Zoom In</u> 	<u>Zoom Out</u> 	<u>Jhenaidah Municipality</u> 
Allows to Zoom In within the map	Allows to Zoom Out within the map	Allow to zoom to Jhenaidah Municipality
<u>Navigate</u> 	<u>Info</u> 	<u>Coordinate Information</u> 
Allow to grab & navigate the map without zoom in/zoom out	Provides the info of the selected item	Shows longitude and latitude of the selected location
<u>Measure Distance</u> 	<u>Measure Area</u> 	<u>Print</u> 
Measure distance of the line drawn	Measure area of the polygon drawn	Prints the visible area of the map
<u>Remove Markers</u> 	<u>Search</u> Landmarks 	<u>Search</u> BIN 
Measure distance of the line drawn	Search places and road by name.	Search buildings by BIN or Holding Number or Tax Code

Fig. 114: General tools available in IMIS Web Map

Info

This tool will detail attribute information of layers at a point clicked on the Map. If there are no data layers at a clicked point, no information will be returned. Information will be displayed at the bottom of the map. There will be a separate information tab for all layers that are turned on in the map.

- Zoom to an area of interest with using navigation tools such Zoom In, Zoom Out & Navigate (Pan)
- Click on **Info** to activate it
- Click on feature on the map of which the user wants to get information.

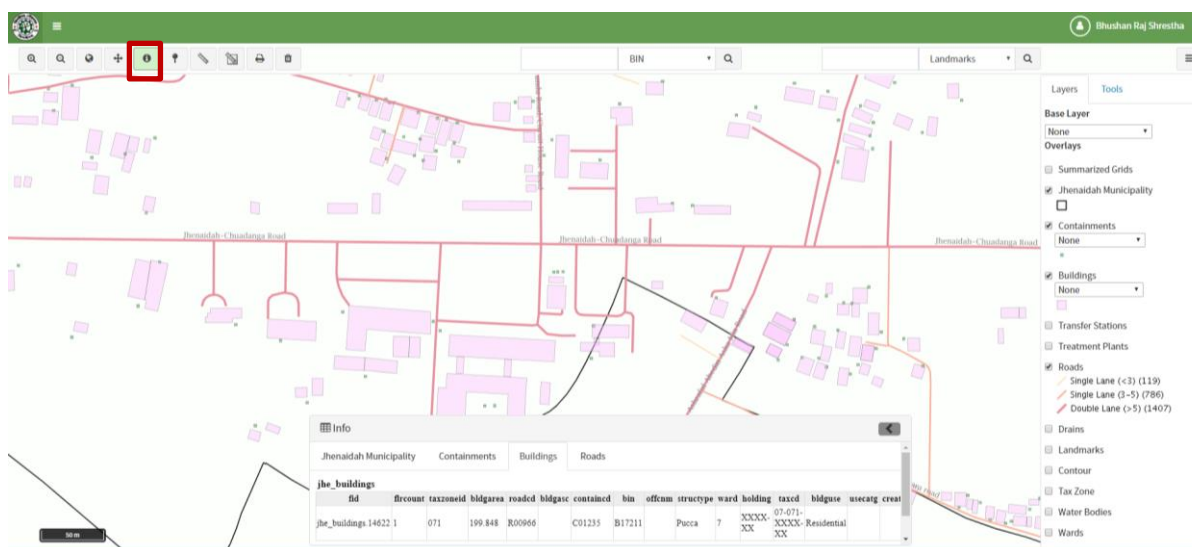


Fig. 115: Getting detail information about a feature in a data layer with **Info** tool

Coordinate Information

This tool will provide a coordinate information of a clicked point on a map. Coordinates information are provided in two coordinate systems (EPSG: 3587 & EPSG: 4326).

- Zoom to an area of interest with using navigation tools such Zoom In, Zoom Out & Navigate (Pan)
- Click on **Coordinate Information** tool to activate it
- Click on a location on the map to get coordinate information of that clicked point
- Click on **Coordinate Information** tool to deactivate it



Fig. 116: Getting coordinate information with **Info** tool

Measure Distance

This tool is useful if the user wants to measure a distance in the map. E.g. the user may want to measure how far a building is from a road.

- Zoom to an area of interest with using navigation tools such Zoom In, Zoom Out & Navigate (Pan)
- Click on **Measure Distance** tool to activate it
- Click on a location on the map (Start Point of a line) and drag the line appeared on the map to another location on the map (End Point line). Distance information is provided in meters. Double Click to finish.
- Click on **Measure Distance** again to exit the tool

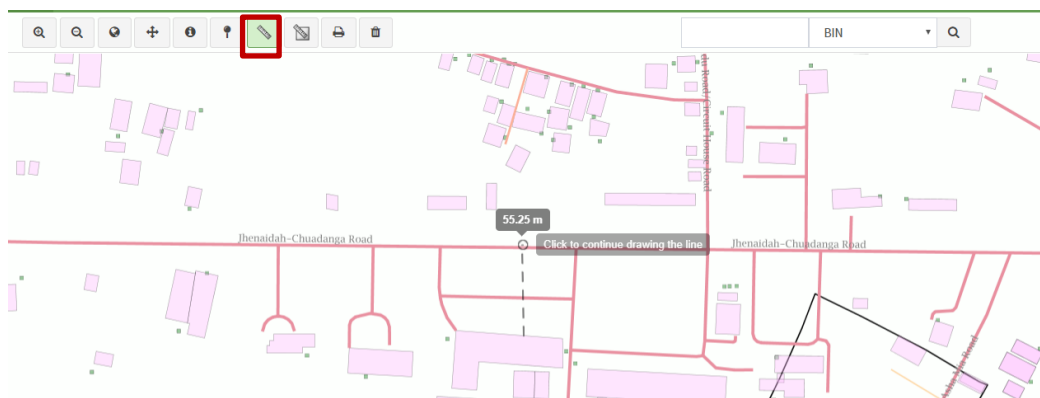


Fig. 117: Measuring distance with Measure Distance tool

Measure Area

This tool is useful if the user wants to measure an area in the map.

- Zoom to an area of interest with using navigation tools such Zoom In, Zoom Out & Navigate (Pan)

- Click on **Measure Area** tool to activate it
- Click on a series of location on the map that represent boundary of an area. Area information is provided in square meters. Double Click to finish.
- Click on **Measure Area** again to exit the tool



Fig. 118: Measuring distance with Measure Area tool

Print

- Arrange a view on the map as desired and turn of/off data layers as desired
- Activate Print tool by clicking it
- A pop up will appear, enter Title and Comments if any on displayed output
- Click **Save**

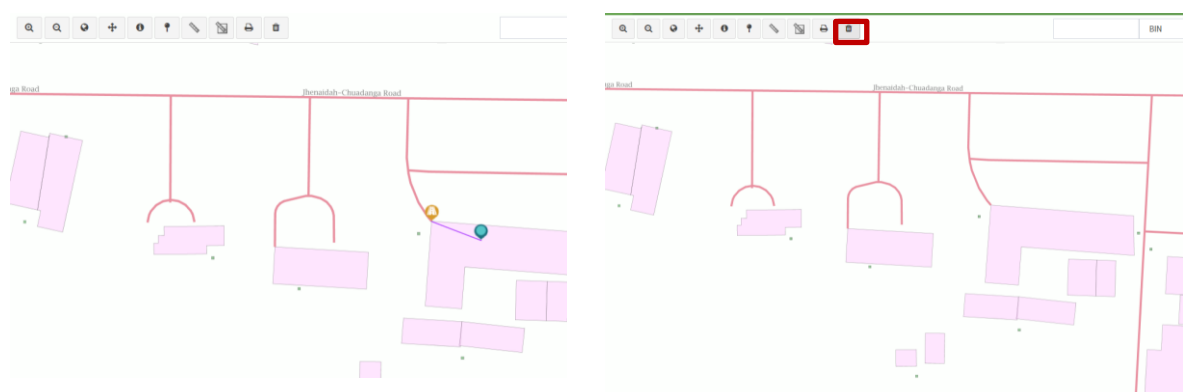


Fig. 119: Print tool

Remove Markers

Some of the tools in Tools TAB (e.g. Find Nearest Road) generators temporary Markers for visualization on the map. This tools is used for removing such Markers.

- Click on **Remove Markers** tool to activate it
- Markers are removed from the map

Fig. 120: Getting rid of temporary Markers with **Remove Markers** tool

Search

If the user wants to navigate to a specific location (E.g. specific road, specific building based on Holding No.) in the map, search tools are extremely beneficial. There are two search tools.

The first search tool provide criteria to search a specific building based on criteria such BIN, Holding No & Tax Code.

- Choose a desired criteria
- Type a Code based on which a building needs to be located in the map
- Click **Search** button
- If a building is found in the database matching the given criteria, the map will be automatically moved to the matched building locations.

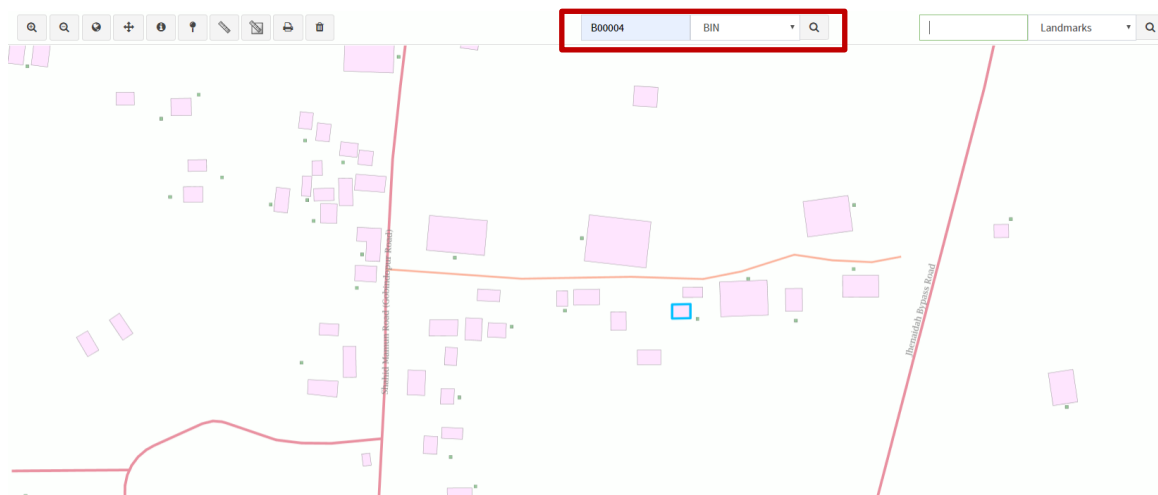


Fig. 121: Searching for a specific building based on BIN

The second search tool provide criteria such as Landmarks & Road Name. The second search tool provides an autocomplete feature to aid the user. The autocomplete feature provides dynamic results that change as the user type text in the search box. Then the user can select one of the results given by the autocomplete feature. E.g. if a user needs to find a particular college, the user can just type college and all matching results will be provided.

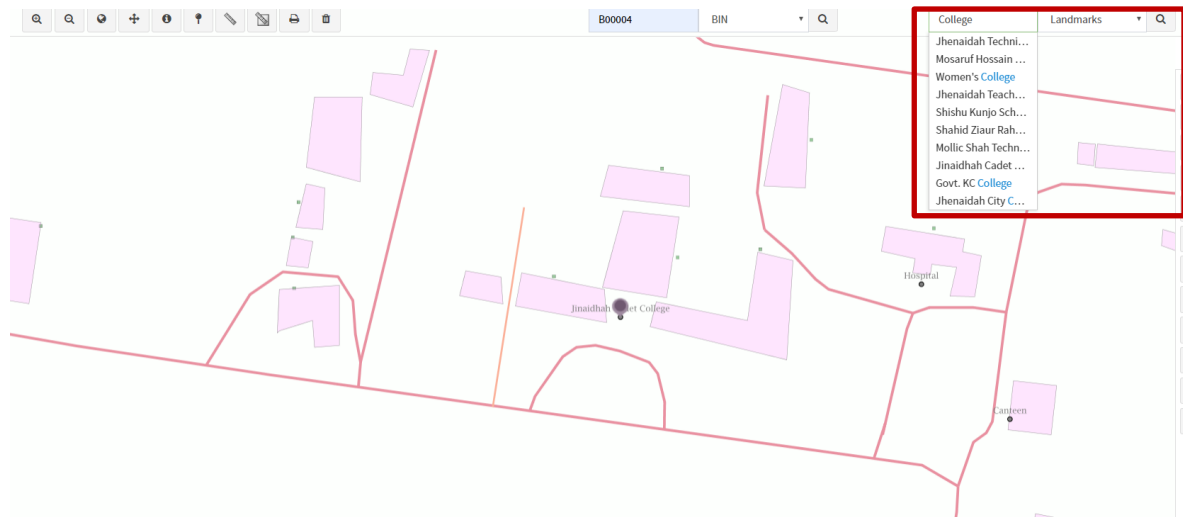


Fig. 122: Searching for a specific building based on Landmarks name

11.2 Layers TAB

Layers TAB enlists all the GIS data layers included in IMIS. For ease of visualization, many base layers (e.g. Google, Bings, OSM) are provided. There is Summarized Grids layer (1 KM grid) that is extremely helpful to get the spatial status of the Municipality based on specific criteria e.g. No. of Buildings. List of data layers shown to a user is governed by their login credentials. Visibility of data layers are controlled by nearby checkbox. Many data layers (Summarized Grids, Containment, and Buildings & Wards) can be visualized based on more than criteria (listed on a dropdown).

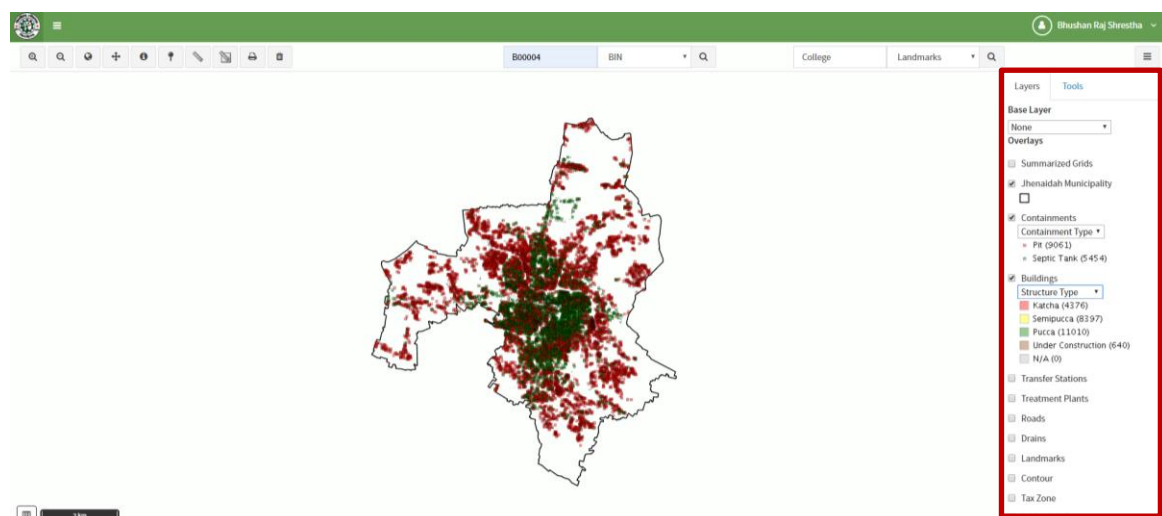


Fig. 123: Data layers available in Data TAB

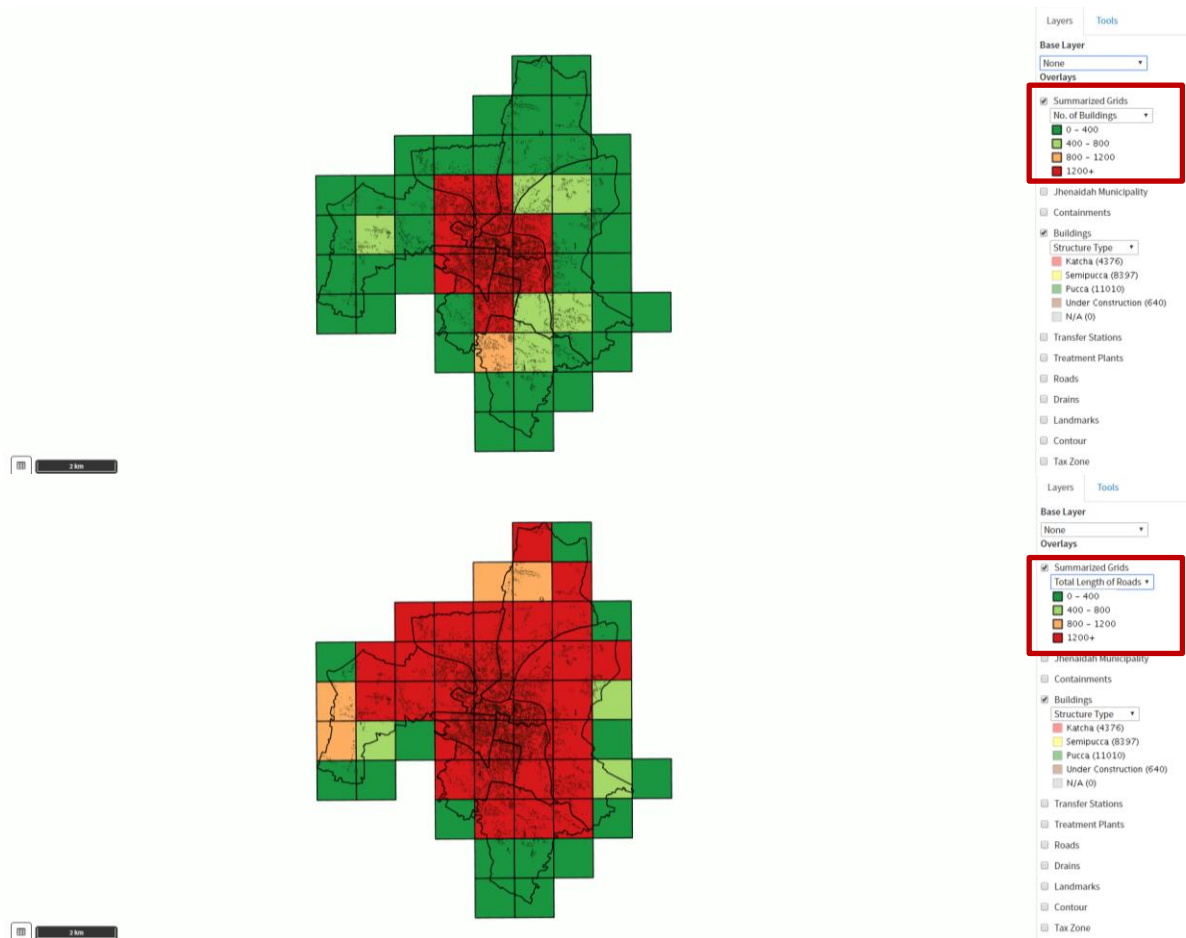


Fig. 124: Visualization of summarized Grids based on two different criteria

- Click nearby checkbox to toggle ON/OFF a specific data layer
- If a data layer has multiple visualization options, a drop down list will appear. Change the options in the list to change visualization based on different attributes of the data layer

11.3 Tools TAB

Tools TAB enlists different specialized tools developed specifically to cater the needs of various users of IMIS. There are all together 13 different tools. The availability of tools will be governed by login credentials of the user.

Table 15: Available tools in Tools TAB and their functions

Tools	Function
 Find Nearest Road	Shows the nearest road to clicked point on the map
 Find Tax Due Buildings	Shows tax due buildings
 Find Applications	Shows applications received for Emptying containments
 Find Containments to ... +	Shows containments to be emptied on the selected date or date range.
 Filter by Wards +	Filter specific data layers by within selected wards
 Filter by Tax Zones +	Filter specific data layers by within selected Tax Zone
 Buildings by Structure ... +	Filter buildings by specific building structure types
 Summary Information	Generates information of building and containment within custom defined boundary
 Export Data Set	Export data layers within custom defined boundary
 Find Buildings to Drain	Shows buildings connected to a specific drain
 Drain Potential Buildings	Shows potential Buildings that have possibility of connecting to a specific drain
 Tax Zone change Impact	Export Building data impacted by possible changes in Tax Zones

11.3.1 Find Nearest Road

- Toggle on building, road, containment layers with an underlying base layer
- Activate the **Find Nearest Road** tool by clicking it
- Click any point on the map. Nearest Road will be shown with a Marker. Use **Info** tool to get information about the Nearest Road, if needed. Use **Measure Distance** tool to get distance information between clicked out and nearest road, if needed.
- Click **Remove Markers** tool to get rid of markers from the map



Fig. 125: Find Nearest Road tool

11.3.2 Find Tax Due Buildings

- Toggle on building layer by clicking associated checkbox
- Activate the **Find Nearest Road** tool by clicking it. Buildings that are due to pay tax will be shown on the map with a green marker.
- Use **Info** tool to get information about due buildings. Click **Remove Markers** tool to get rid of markers from the map



Fig. 126: Find Tax Due Buildings tool

11.3.3 Find Application

- Toggle on containment layer by clicking associated checkbox
- Activate the **Find Application** tool by clicking it. Applications received for emptying service request will be shown on the map with markers with varied colors according to its status.

Table 16: Applications markers according to its status

Red Markers	Just in Applications
Blue Markers	Assessment done
Yellow Markers	Emptying done
Green Markers	Feedback collected

- Use **Info** tool to get information about buildings. Use **Find Nearest Road** tool to find nearest road. Use **Measure Distance** tool to get idea about distance to the nearest road.
- Click **Remove Markers** tool to get rid of markers from the map.



Fig. 127: Find Application tool

11.3.4 Find Containment to be emptied

- Toggle on containment layer by clicking associated checkbox
- Activate the **Find Containment to be emptied** tool by clicking it. There are different options
 - Next X days - if the user wants to find potential containment for emptying for coming X days

- Next Week - if the user wants to find potential containment for emptying in Next Week
- Date – if the user wants to find potential containment for emptying in a specific date
- Choose an option as desired and then click search
- Potential containment to be emptied will be shown on the map with a green marker and total number of potential containment will be given in a dialogue box.
- Click **Remove Markers** tool to get rid of markers from the map.

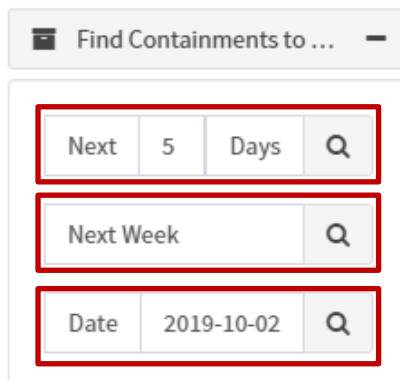


Fig. 128: Options for finding potential containment to be emptied tool

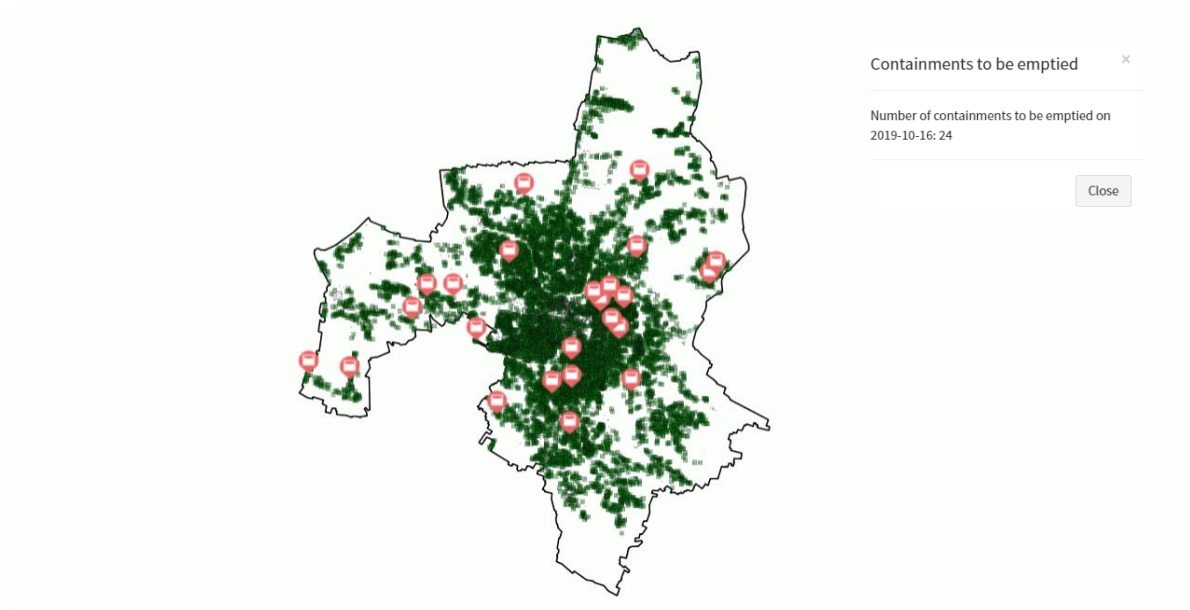


Fig. 129: Find containment to be emptied tool

11.3.5 Filter by Wards

- Activate the **Filter by Wards** tool by clicking it
- Select the wards that the user wanted to confine data export. Multiple wards can be selected.
- Choose data layers desired by the used for export. Multiple data layers can be selected.
- Click **Filter** button to filter data. The map will be updated.
- Choose desired format (JSON, KML, Shape File) under **Export** drop down
- Once the user is done with export, click **Clear** button to remove applied filter

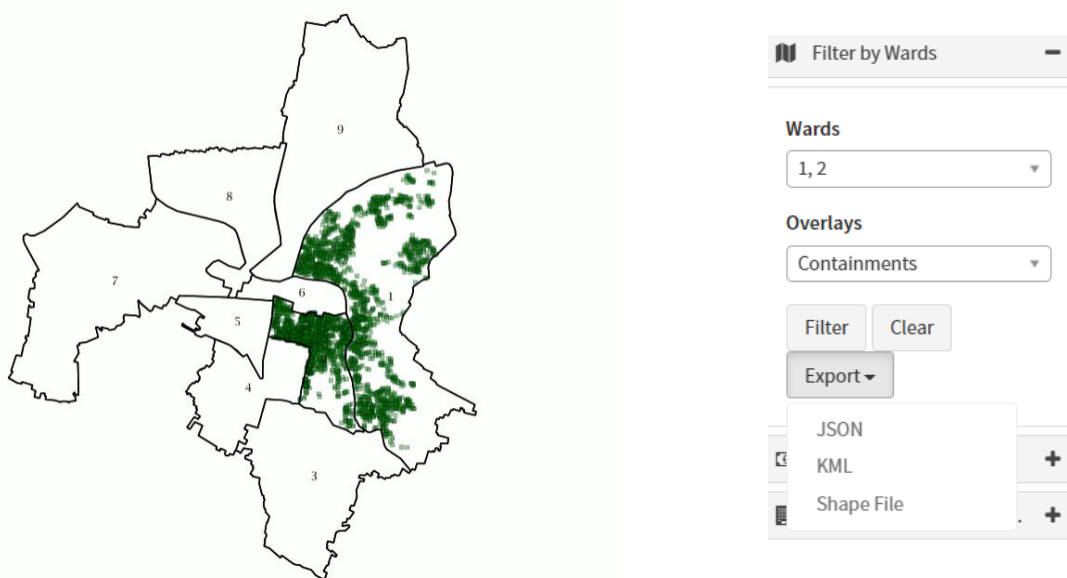


Fig. 130: Filter by Ward tool

11.3.6 Filter by Tax Zones

- Activate the **Filter by Tax Zones** tool by clicking it
- Select the Tax Zones that the user wanted to confine data export. Multiple Tax Zones can be selected.
- Choose data layers desired by the used for export. Multiple data layers can be selected.
- Click **Filter** button to filter data. The map will be updated.
- Choose desired format (JSON, KML, Shape File) under **Export** drop down
- Once the user is done with export, click **Clear** button to remove applied filter

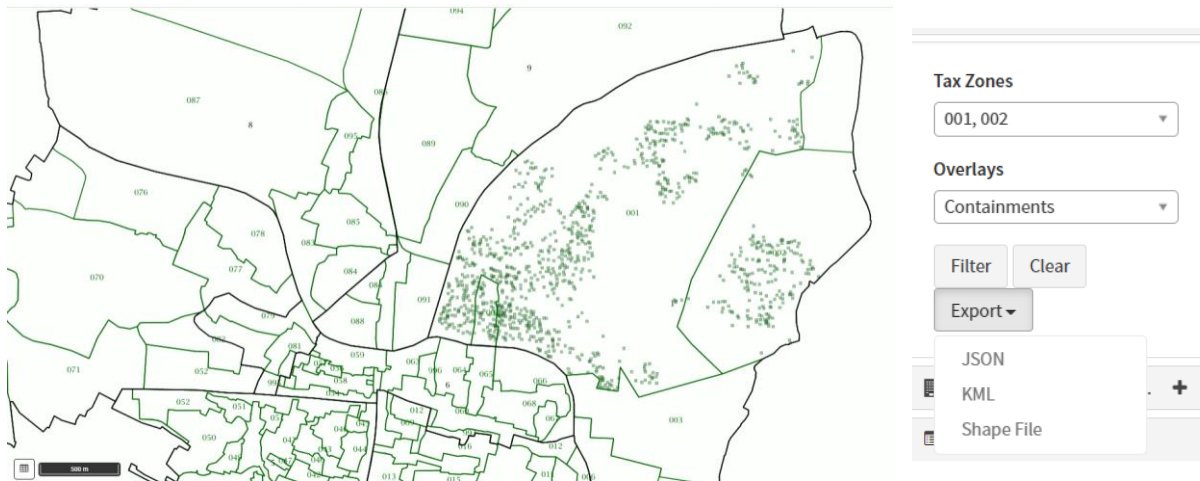


Fig. 131: Filter by Tax Zones tool

11.3.7 Buildings by Structures Type

- Toggle on Building layer by clicking associated checkbox
- Activate the **Buildings by Structure Type** tool by clicking it
- Select desired structure types options among Kachha, Semipucca, Pucca & Under Construction. The options can be combined.
- The map is instantly updated
- Toggle Off applied structure types options to remove applied filter

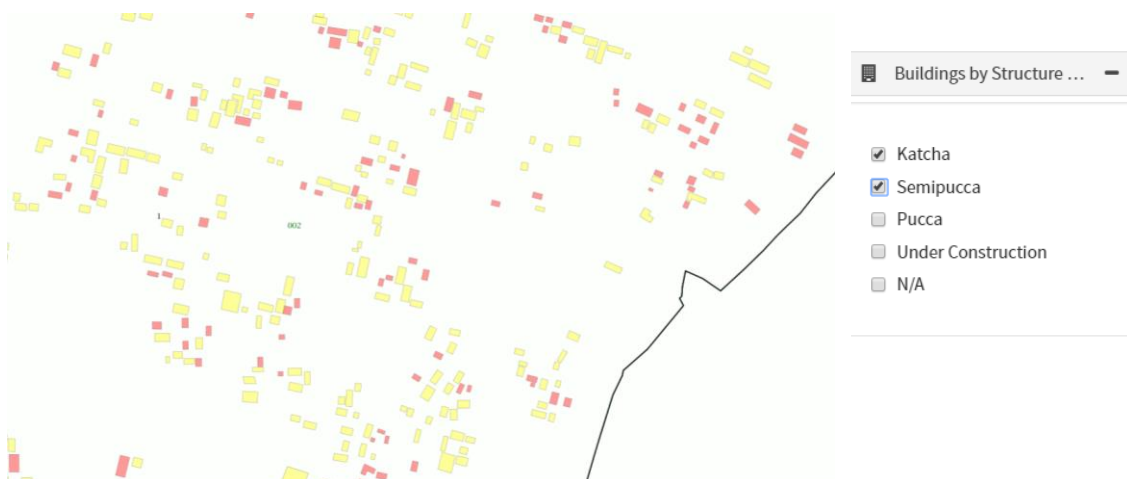


Fig. 132: Building by Structure Type tool

11.3.8 Summary Information

- Toggle on Building & Containment layer by clicking associated checkbox

- Activate the **Summary Information** tool by clicking it
- Click on maps to make a custom boundary. Double click to finish
- A dialogue box will pop that will provide summary information about building and containments with the custom boundary
- Close the dialogue box
- Click **Remove Markers** tool to get rid of markers from the map
- Deactivate the **Summary Information** tool by clicking it again



Fig. 133: Summary Information tool

11.3.9 Export Data Set

- Activate the **Export Data Set** tool by clicking it
- Click on maps to make a custom boundary. Double click to finish
- A dialogue box will pop asking for datasets to be exported. Pick desired datasets for export
- Click on desired format (JSON, KML, Shape File) for export
- Close the dialogue box
- Deactivate the **Export Data Set** tool by clicking it again



Fig. 134: Export Data Set tool

11.3.10 Find Building to Drain

- Toggle on Building & Drain layer by clicking associated checkbox
- Activate the **Find Building to Drain** tool by clicking it
- Select a drain by clicking on a drain
- If some buildings are connected to clicked drain, surrounding buildings will be highlighted in yellow
- Deactivate the **Find Building to Drain** tool by clicking it again

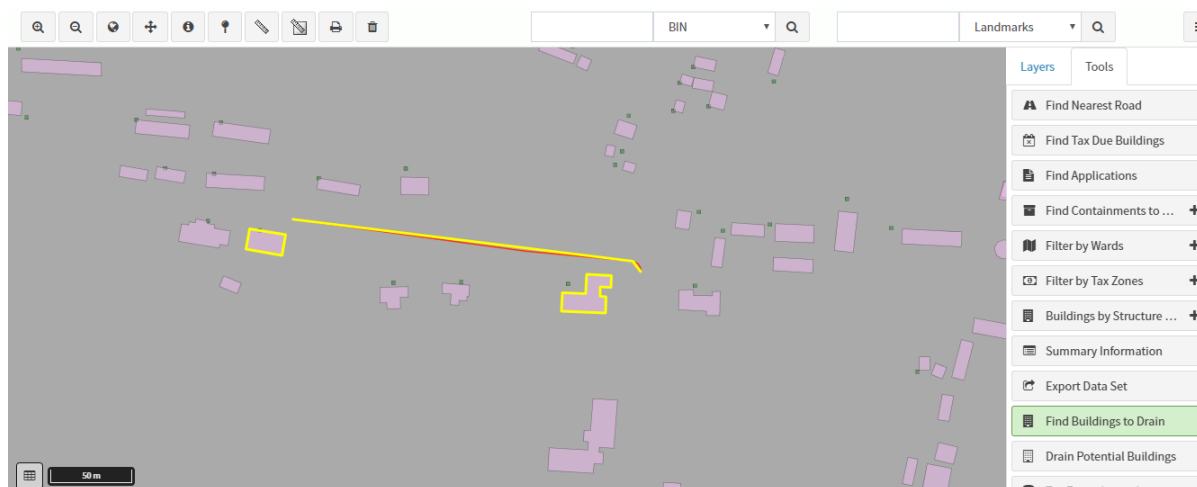


Fig. 135: Find Building to Drain Tool

11.3.11 Drain Potential Building

- Toggle on Building & Drain layer by clicking associated checkbox
- Activate the **Find Building to Drain** tool by clicking it
- Select a drain by clicking on a drain. The user is asked for a buffer distance in a dialogue box. Provide distance in meter and click **Get Buildings** button
- Buildings that are within the provided buffer distance will be highlighted in yellow
- If some buildings are connected to clicked drain, surrounding buildings will highlighted in yellow
- Deactivate the **Drain Potential Building** tool by clicking it again

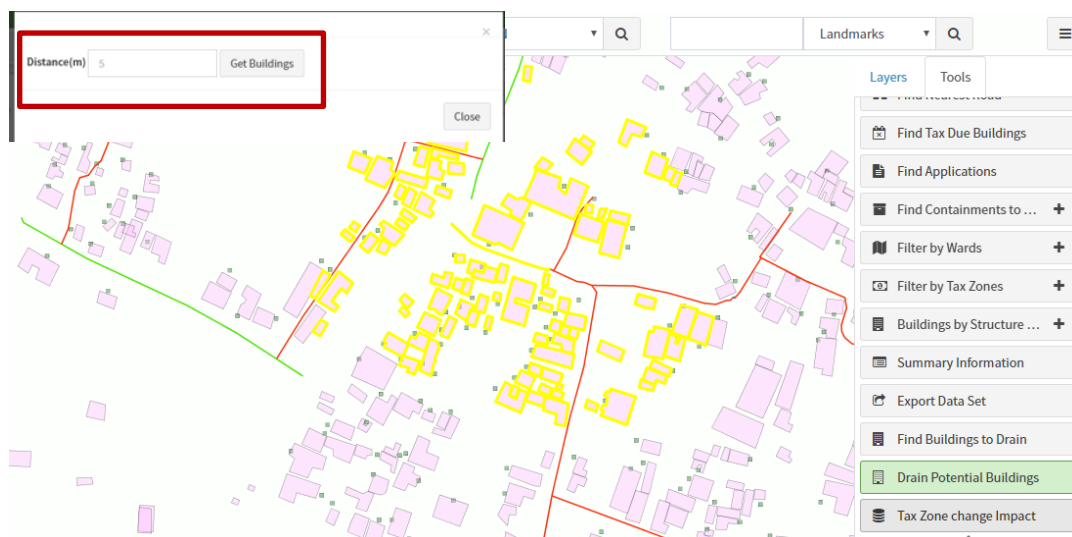


Fig. 136: Drain Potential Buildings tool

11.3.12 Tax Zone Change Impact

- Toggle on Building & Drain layer by clicking associated checkbox
- Activate the **Tax Zone Change Impact** tool by clicking it
- Click on maps to make a custom boundary. Double click to finish
- A dialogue box will pop. Click CSV button
- Close the dialogue box
- Deactivate the **Tax Zone Change Impact** tool by clicking it



Fig. 137: Tax Zone Change Impact

IMIS FRAMEWORK

IMIS stands for **Integrated GIS-based Municipal Information System** developed for managing faecal sludge in Khulna City Corporation



KDA Detailed Area Plan (GIS-2014)

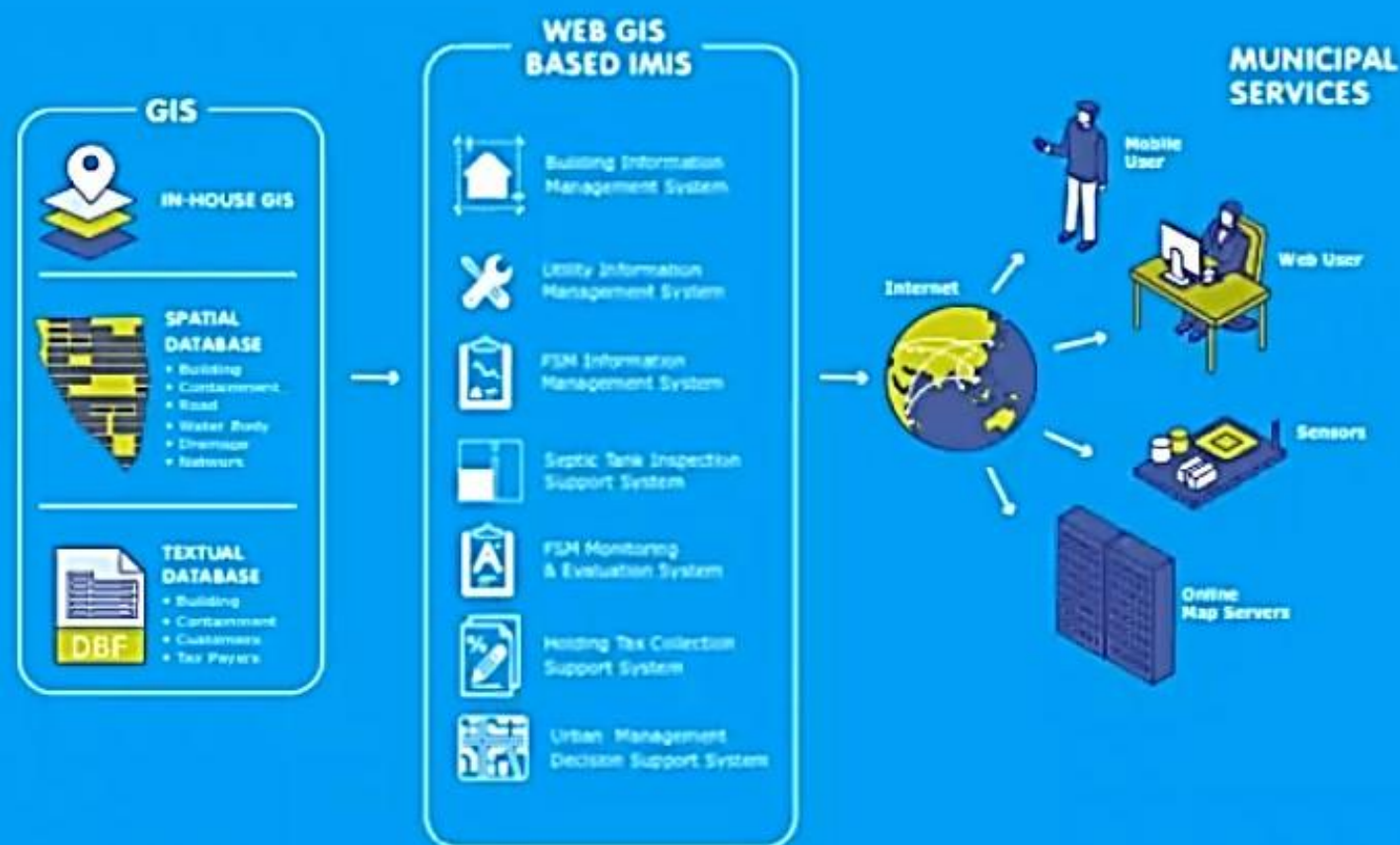
MAPS

- In house Data
- Spatial Database



Jhenaidah Municipality Information Framework (IMIS)

IMIS enables information based decision making and planning for **Municipal Services** in line with **SMART City** initiatives



IMIS is capable of providing city level data. For example:

